

Basic RL and RC Circuits

INTRODUCTION

In Chap. 7 we wrote equations for the response of several circuits containing both inductance and capacitance, but we did not solve any of them. Now we are ready to proceed with the solution of the simpler circuits, namely, those which contain only resistors and inductors, or only resistors and capacitors.

Although the circuits we are about to consider have a very elementary appearance, they are also of practical importance. Networks of this form find use in electronic amplifiers, automatic control systems, operational amplifiers, communications equipment, and many other applications. Familiarity with these simple circuits will enable us to predict the accuracy with which the output of an amplifier can follow an input that is changing rapidly with time, or to predict how quickly the speed of a motor will change in response to a change in its field current. Our understanding of simple RL and RC circuits will also enable us to suggest modifications to the amplifier or motor in order to obtain a more desirable response.

8.1 THE SOURCE-FREE RL CIRCUIT

The analysis of circuits containing inductors and/or capacitors is dependent upon the formulation and solution of the integrodifferential equations that characterize the circuits. We will call the special type of equation we obtain a *homogeneous linear differential equation*, which is simply a differential equation in which every term is of the first degree in the dependent variable or one of its derivatives. A solution is obtained when we have found an expression for the

KEY CONCEPTS

RL and RC Time Constants

Natural and Forced Response

Calculating the Time-Dependent Response to DC Excitation

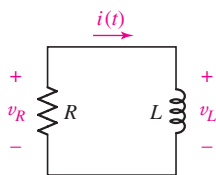
How to Determine Initial Conditions and Their Effect on the Circuit Response

Analyzing Circuits with Step Function Input and with Switches

Construction of Pulse Waveforms Using Unit-Step Functions

The Response of Sequentially Switched Circuits





■ FIGURE 8.1 A series RL circuit for which $i(t)$ is to be determined, subject to the initial condition that $i(0) = I_0$.

It may seem pretty strange to discuss a time-varying current flowing in a circuit with no sources! Keep in mind that we only know the current at the time specified as $t = 0$; we don't know the current prior to that time. In the same vein, we don't know what the circuit looked like prior to $t = 0$, either. In order for a current to be flowing, a source had to have been present at some point, but we are not privy to this information. Fortunately, it is not required in order to analyze the circuit we are given.

dependent variable that satisfies both the differential equation and also the prescribed energy distribution in the inductors or capacitors at a prescribed instant of time, usually $t = 0$.

The solution of the differential equation represents a response of the circuit, and it is known by many names. Since this response depends upon the general “nature” of the circuit (the types of elements, their sizes, the interconnection of the elements), it is often called a **natural response**. However, any real circuit we construct cannot store energy forever; the resistances intrinsically associated with inductors and capacitors will eventually convert all stored energy into heat. The response must eventually die out, and for this reason it is frequently referred to as the **transient response**. Finally, we should also be familiar with the mathematicians’ contribution to the nomenclature; they call the solution of a homogeneous linear differential equation a *complementary function*.

When we consider independent sources acting on a circuit, part of the response will resemble the nature of the particular source (or *forcing function*) used; this part of the response, called the *particular solution*, the *steady-state response*, or the **forced response**, will be “complemented” by the complementary response produced in the source-free circuit. The complete response of the circuit will then be given by the sum of the complementary function and the particular solution. In other words, the complete response is the sum of the natural response and the forced response. The source-free response may be called the *natural response*, the *transient response*, the *free response*, or the *complementary function*, but because of its more descriptive nature, we will most often call it the *natural response*.

We will consider several different methods of solving these differential equations. The mathematical manipulation, however, is not circuit analysis. Our greatest interest lies in the solutions themselves, their meaning, and their interpretation, and we will try to become sufficiently familiar with the form of the response that we are able to write down answers for new circuits by just plain thinking. Although complicated analytical methods are needed when simpler methods fail, a well-developed intuition is an invaluable resource in such situations.

We begin our study of transient analysis by considering the simple series RL circuit shown in Fig. 8.1. Let us designate the time-varying current as $i(t)$; we will represent the value of $i(t)$ at $t = 0$ as I_0 ; in other words, $i(0) = I_0$. We therefore have

$$Ri + v_L = Ri + L \frac{di}{dt} = 0$$

or

$$\frac{di}{dt} + \frac{R}{L}i = 0 \quad [1]$$

Our goal is an expression for $i(t)$ which satisfies this equation *and* also has the value I_0 at $t = 0$. The solution may be obtained by several different methods.

A Direct Approach

One very direct method of solving a differential equation consists of writing the equation in such a way that the variables are separated, and then

integrating each side of the equation. The variables in Eq. [1] are i and t , and it is apparent that the equation may be multiplied by dt , divided by i , and arranged with the variables separated:

$$\frac{di}{i} = -\frac{R}{L} dt \quad [2]$$

Since the current is I_0 at $t = 0$ and $i(t)$ at time t , we may equate the two definite integrals which are obtained by integrating each side between the corresponding limits:

$$\int_{I_0}^{i(t)} \frac{di'}{i'} = \int_0^t -\frac{R}{L} dt'$$

Performing the indicated integration,

$$\ln i' \Big|_{I_0}^i = -\frac{R}{L} t' \Big|_0^t$$

which results in

$$\ln i - \ln I_0 = -\frac{R}{L}(t - 0)$$

After a little manipulation, we find that the current $i(t)$ is given by

$$i(t) = I_0 e^{-Rt/L} \quad [3]$$

We check our solution by first showing that substitution of Eq. [3] in Eq. [1] yields the identity $0 = 0$, and then showing that substitution of $t = 0$ in Eq. [3] produces $i(0) = I_0$. Both steps are necessary; the solution must satisfy the differential equation which characterizes the circuit, and it must also satisfy the initial condition.



EXAMPLE 8.1

If the inductor of Fig. 8.2 has a current $i_L = 2$ A at $t = 0$, find an expression for $i_L(t)$ valid for $t > 0$, and its value at $t = 200 \mu\text{s}$.

This is the identical type of circuit just considered, so we expect an inductor current of the form

$$i_L = I_0 e^{-Rt/L}$$

where $R = 200 \Omega$, $L = 50$ mH and I_0 is the initial current flowing through the inductor at $t = 0$. Thus,

$$i_L(t) = 2e^{-4000t}$$

Substituting $t = 200 \times 10^{-6}$ s, we find that $i_L(t) = 898.7$ mA, less than half the initial value.

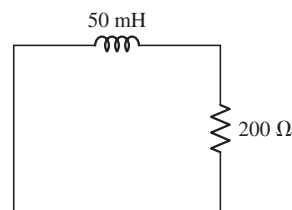


FIGURE 8.2 A simple RL circuit in which energy is stored in the inductor at $t = 0$.

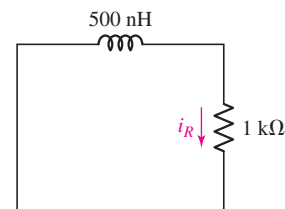


FIGURE 8.3 Circuit for Practice Problem 8.1.

PRACTICE

8.1 Determine the current i_R through the resistor of Fig. 8.3 at $t = 1$ ns if $i_R(0) = 6$ A.

Ans: 812 mA.

An Alternative Approach

The solution may also be obtained by a slight variation of the method we just described. After separating the variables, we now also include a constant of integration. Thus,

$$\int \frac{di}{i} = - \int \frac{R}{L} dt + K$$

and integration gives us

$$\ln i = -\frac{R}{L}t + K \quad [4]$$

The constant K cannot be evaluated by substitution of Eq. [4] in the original differential equation [1]; the identity $0 = 0$ will result, because Eq. [4] is a solution of Eq. [1] for *any* value of K (try it out on your own). The constant of integration must be selected to satisfy the initial condition $i(0) = I_0$. Thus, at $t = 0$, Eq. [4] becomes

$$\ln I_0 = K$$

and we use this value for K in Eq. [4] to obtain the desired response

$$\ln i = -\frac{R}{L}t + \ln I_0$$

or

$$i(t) = I_0 e^{-Rt/L}$$

as before.

A More General Solution Approach

Either of these methods can be used when the variables are separable, but this is not always the situation. In the remaining cases we will rely on a very powerful method, the success of which will depend upon our intuition or experience. We simply guess or assume a form for the solution and then test our assumptions, first by substitution in the differential equation, and then by applying the given initial conditions. Since we cannot be expected to guess the exact numerical expression for the solution, we will assume a solution containing several unknown constants and select the values for these constants in order to satisfy the differential equation and the initial conditions. Many of the differential equations encountered in circuit analysis have a solution which may be represented by the exponential function or by the sum of several exponential functions. Let us assume a solution of Eq. [1] in exponential form,

$$i(t) = Ae^{s_1 t} \quad [5]$$

where A and s_1 are constants to be determined. After substituting this assumed solution in Eq. [1], we have

$$As_1 e^{s_1 t} + A \frac{R}{L} e^{s_1 t} = 0$$

or

$$\left(s_1 + \frac{R}{L}\right) Ae^{s_1 t} = 0 \quad [6]$$

In order to satisfy this equation for all values of time, it is necessary that $A = 0$, or $s_1 = -\infty$, or $s_1 = -R/L$. But if $A = 0$ or $s_1 = -\infty$, then every response is zero; neither can be a solution to our problem. Therefore, we must choose

$$s_1 = -\frac{R}{L} \quad [7]$$

and our assumed solution takes on the form

$$i(t) = Ae^{-Rt/L}$$

The remaining constant must be evaluated by applying the initial condition $i(0) = I_0$. Thus, $A = I_0$, and the final form of the assumed solution is (again)

$$i(t) = I_0 e^{-Rt/L}$$

A summary of the basic approach is outlined in Fig. 8.4.

A Direct Route: The Characteristic Equation

In fact, there is a more direct route that we can take. In obtaining Eq. [7], we solved

$$s_1 + \frac{R}{L} = 0 \quad [8]$$

which is known as the **characteristic equation**. We can obtain the characteristic equation directly from the differential equation, without the need for substitution of our trial solution. Consider the general first-order differential equation

$$a \frac{df}{dt} + bf = 0$$

where a and b are constants. We substitute s^1 for df/dt and s^0 for f , resulting in

$$a \frac{df}{dt} + bf = (as + b)f = 0$$

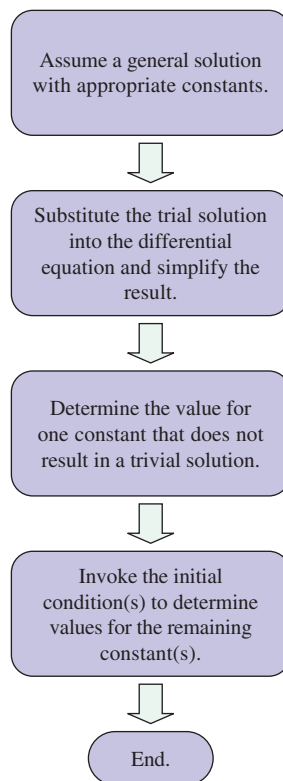
From this we may directly obtain the characteristic equation

$$as + b = 0$$

which has the single root $s = -b/a$. The solution to our differential equation is then

$$f = Ae^{-bt/a}$$

This basic procedure is easily extended to second-order differential equations, as we will explore in Chap. 9.



■ FIGURE 8.4 Flowchart for the general approach to solution of first-order differential equations where, based on experience, we can guess the form of the solution.

EXAMPLE 8.2

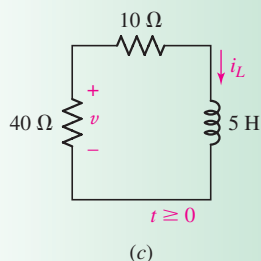
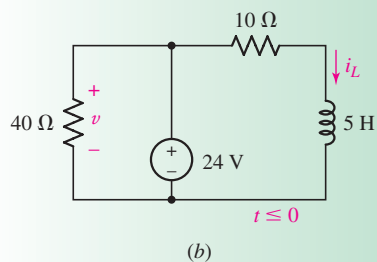
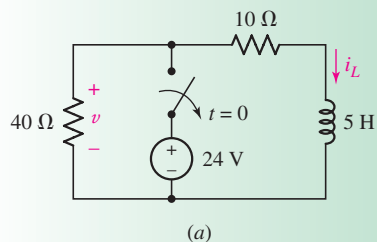


FIGURE 8.5 (a) A simple RL circuit with a switch thrown at time $t = 0$. (b) The circuit as it exists prior to $t = 0$. (c) The circuit after the switch is thrown, and the 24 V source is removed.



For the circuit of Fig. 8.5a, find the voltage labeled v at $t = 200$ ms.

► **Identify the goal of the problem.**

The schematic of Fig. 8.5a actually represents *two different* circuits: one with the switch closed (Fig. 8.5b) and one with the switch open (Fig. 8.5c). We are asked to find $v(0.2)$ for the circuit shown in Fig. 8.5c.

► **Collect the known information.**

Both new circuits are drawn and labeled correctly. We next make the assumption that the circuit in Fig. 8.5b has been connected for a long time, so that any transients have dissipated. We may make such an assumption as a general rule unless instructed otherwise. This circuit determines $i_L(0)$.

► **Devise a plan.**

The circuit of Fig. 8.5c may be analyzed by writing a KVL equation. Ultimately we want a differential equation with only v and t as variables; we will then solve the differential equation for $v(t)$.

► **Construct an appropriate set of equations.**

Referring to Fig. 8.5c, we write

$$-v + 10i_L + 5 \frac{di_L}{dt} = 0$$

Substituting $i_L = -v/40$, we find that

$$\frac{5}{40} \frac{dv}{dt} + \left(\frac{10}{40} + 1 \right) v = 0$$

or, more simply,

$$\frac{dv}{dt} + 10v = 0 \quad [9]$$

► **Determine if additional information is required.**

From previous experience, we know that a complete expression for v will require knowledge of v at a specific instant of time, with $t = 0$ being the most convenient. We might be tempted to look at Fig. 8.5b and write $v(0) = 24$ V, but this is only true *just before the switch opens*. The resistor voltage can change to any value in the instant that the switch is thrown; only the inductor current must remain unchanged.

In the circuit of Fig. 8.5b, $i_L = 24/10 = 2.4$ A since the inductor acts like a short circuit to a dc current. Therefore, $i_L(0) = 2.4$ A in the circuit of Fig. 8.5c, as well—a key point in analyzing this type of circuit. Therefore, in the circuit of Fig. 8.5c, $v(0) = (40)(-2.4) = -96$ V.

► **Attempt a solution.**

Any of the three basic solution techniques can be brought to bear; let's start by writing the characteristic equation corresponding to Eq. [9]:

$$s + 10 = 0$$

Solving, we find that $s = -10$, so

$$v(t) = Ae^{-10t} \quad [10]$$

(which, upon substitution into the left-hand side of Eq. [9], results in

$$-10Ae^{-10t} + 10Ae^{-10t} = 0$$

as expected.)

We find A by setting $t = 0$ in Eq. [10] and employing the fact that $v(0) = -96$ V. Thus,

$$v(t) = -96e^{-10t} \quad [11]$$

and so $v(0.2) = -12.99$ V, down from a maximum of -96 V.

► **Verify the solution. Is it reasonable or expected?**

Instead of writing a differential equation in v , we could have written our differential equation in terms of i_L :

$$40i_L + 10i_L + 5 \frac{di_L}{dt} = 0$$

or

$$\frac{di_L}{dt} + 10i_L = 0$$

which has the solution $i_L = Be^{-10t}$. With $i_L(0) = 2.4$, we find that $i_L(t) = 2.4e^{-10t}$. Since $v = -40i_L$, we once again obtain Eq. [11]. We should note: it is **no coincidence** that the inductor current and the resistor voltage have the same exponential dependence!

PRACTICE

8.2 Determine the inductor voltage v in the circuit of Fig. 8.6 for $t > 0$.

Ans: $-25e^{-2t}$ V.

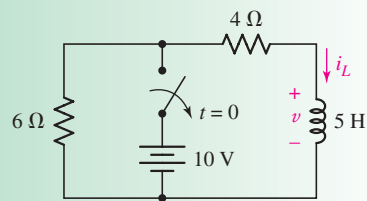


FIGURE 8.6 Circuit for Practice Problem 8.2.

Accounting for the Energy

Before we turn our attention to the interpretation of the response, let us return to the circuit of Fig. 8.1, and check the power and energy relationships. The power being dissipated in the resistor is

$$p_R = i^2 R = I_0^2 R e^{-2Rt/L}$$

and the total energy turned into heat in the resistor is found by integrating the instantaneous power from zero time to infinite time:

$$\begin{aligned} w_R &= \int_0^\infty p_R dt = I_0^2 R \int_0^\infty e^{-2Rt/L} dt \\ &= I_0^2 R \left(\frac{-L}{2R} \right) e^{-2Rt/L} \Big|_0^\infty = \frac{1}{2} L I_0^2 \end{aligned}$$

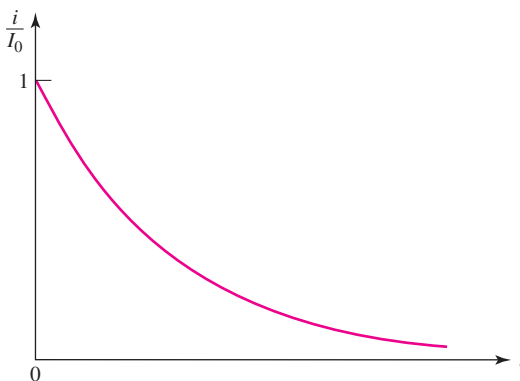
This is the result we expect, because the total energy stored initially in the inductor is $\frac{1}{2} L I_0^2$, and there is no longer any energy stored in the inductor at infinite time since its current eventually drops to zero. All the initial energy therefore is accounted for by dissipation in the resistor.

8.2 PROPERTIES OF THE EXPONENTIAL RESPONSE

Let us now consider the nature of the response in the series RL circuit. We have found that the inductor current is represented by

$$i(t) = I_0 e^{-Rt/L}$$

At $t = 0$, the current has value I_0 , but as time increases, the current decreases and approaches zero. The shape of this decaying exponential is seen by the plot of $i(t)/I_0$ versus t shown in Fig. 8.7. Since the function we are plotting is $e^{-Rt/L}$, the curve will not change if R/L remains unchanged. Thus, the same curve must be obtained for every series RL circuit having the same L/R ratio. Let us see how this ratio affects the shape of the curve.



■ FIGURE 8.7 A plot of $e^{-Rt/L}$ versus t .

If we double the ratio of L to R , the exponent will be unchanged only if t is also doubled. In other words, the original response will occur at a later time, and the new curve is obtained by moving each point on the original curve twice as far to the right. With this larger L/R ratio, the current takes longer to decay to any given fraction of its original value. We might have a tendency to say that the “width” of the curve is doubled, or that the width is proportional to L/R . However, we find it difficult to define our term *width*, because each curve extends from $t = 0$ to ∞ ! Instead, let us consider the time that would be required for the current to drop to zero *if it continued to drop at its initial rate*.

The initial rate of decay is found by evaluating the derivative at zero time:

$$\left. \frac{d}{dt} \frac{i}{I_0} \right|_{t=0} = -\frac{R}{L} e^{-Rt/L} \Big|_{t=0} = -\frac{R}{L}$$

We designate the value of time it takes for i/I_0 to drop from unity to zero, assuming a constant rate of decay, by the Greek letter τ (tau). Thus,

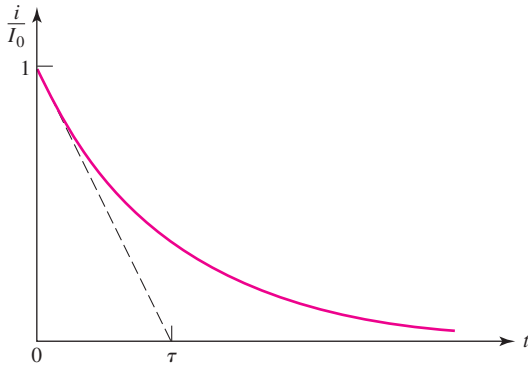
$$\left(\frac{R}{L} \right) \tau = 1$$

or

$$\tau = \frac{L}{R}$$

[12]

The ratio L/R has the units of seconds, since the exponent $-Rt/L$ must be dimensionless. This value of time τ is called the **time constant** and is shown pictorially in Fig. 8.8. The time constant of a series RL circuit may be found graphically from the response curve; it is necessary only to draw the tangent to the curve at $t = 0$ and determine the intercept of this tangent line with the time axis. This is often a convenient way of approximating the time constant from the display on an oscilloscope.

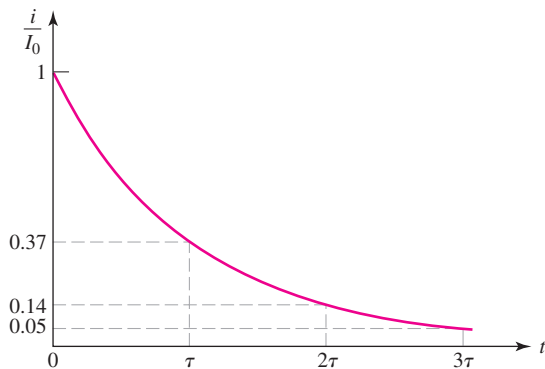


■ FIGURE 8.8 The time constant τ is L/R for a series RL circuit. It is the time required for the response curve to drop to zero if it decays at a constant rate equal to its initial rate of decay.

An equally important interpretation of the time constant τ is obtained by determining the value of $i(t)/I_0$ at $t = \tau$. We have

$$\frac{i(\tau)}{I_0} = e^{-1} = 0.3679 \quad \text{or} \quad i(\tau) = 0.3679I_0$$

Thus, in one time constant the response has dropped to 36.8 percent of its initial value; the value of τ may also be determined graphically from this fact, as indicated by Fig. 8.9. It is convenient to measure the decay of the current at intervals of one time constant, and recourse to a hand calculator shows that $i(t)/I_0$ is 0.3679 at $t = \tau$, 0.1353 at $t = 2\tau$, 0.04979 at $t = 3\tau$, 0.01832 at $t = 4\tau$, and 0.006738 at $t = 5\tau$. At some point three to five time constants after zero time, most of us would agree that the current is a



■ FIGURE 8.9 The current in a series RL circuit is reduced to 37 percent of its initial value at $t = \tau$, 14 percent at $t = 2\tau$, and 5 percent at $t = 3\tau$.



negligible fraction of its former self. Thus, if we are asked, “How long does it take for the current to decay to zero?” our answer might be, “About five time constants.” At that point, the current is less than 1 percent of its original value!

PRACTICE

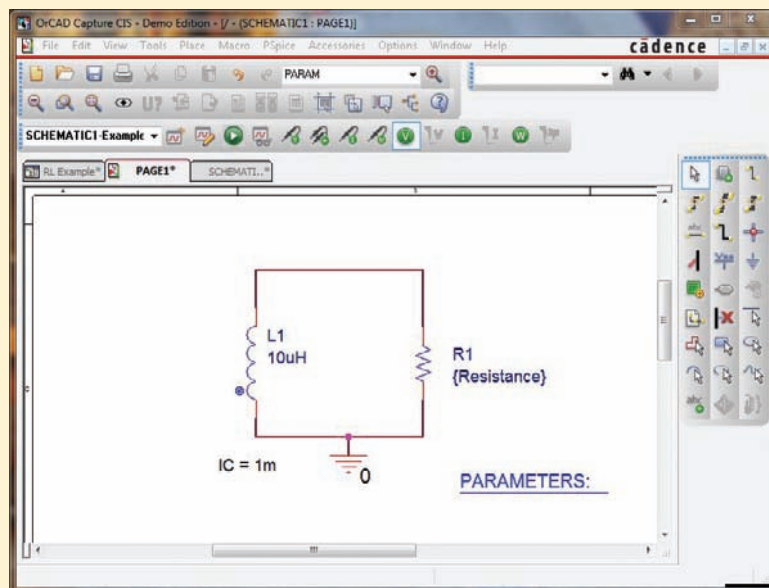
8.3 In a source-free series RL circuit, find the numerical value of the ratio: (a) $i(2\tau)/i(\tau)$; (b) $i(0.5\tau)/i(0)$; (c) t/τ if $i(t)/i(0) = 0.2$; (d) t/τ if $i(0) - i(t) = i(0) \ln 2$.

Ans: 0.368; 0.607; 1.609; 1.181.

COMPUTER-AIDED ANALYSIS

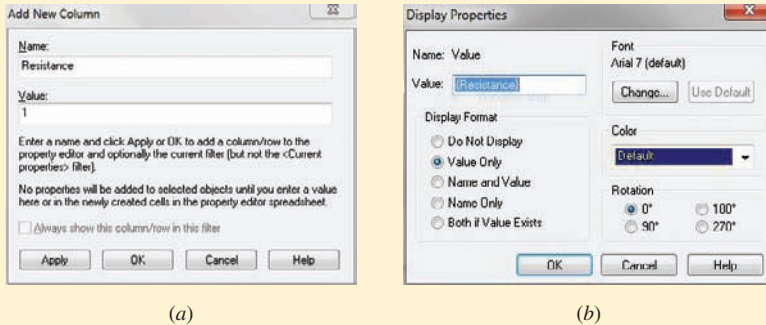
The transient analysis capability of PSpice is very useful when considering the response of source-free circuits. In this example, we make use of a special feature that allows us to vary a component parameter, similar to the way we varied the dc voltage in other simulations. We do this by adding the component **PARAM** to our schematic; it may be placed anywhere, as we will not wire it into the circuit. Our complete RL circuit is shown in Fig. 8.10, which includes an initial inductor current of 1 mA.

In order to relate our resistor value to the proposed parameter sweep, we must perform three tasks. First, we provide a name for our parameter, which we choose to call Resistance for the sake of simplicity. This is accomplished by double-clicking on the **PARAMETERS:** label in the schematic, which opens the Property Editor for this pseudo-component. Clicking on **New Column** results in the dialog box shown in Fig. 8.11a, in which we enter *Resistance* under **Name** and a placeholder value of 1 under **Value**. Our second task consists of linking the



■ FIGURE 8.10 Simple RL circuit drawn using the schematic capture tool.

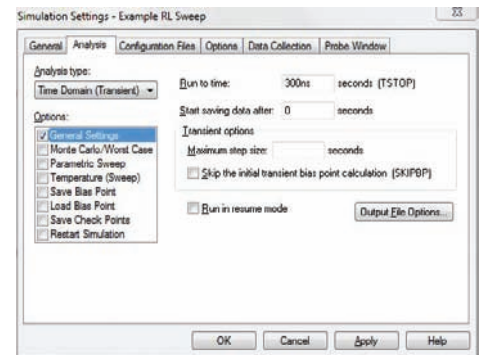
value of R1 to our parameter sweep, which we accomplish by double-clicking on the default value of R1 on the schematic, resulting in the dialog box of Fig. 8.11b. Under **Value**, we simply enter $\{Resistance\}$. (Note the curly brackets are required.)



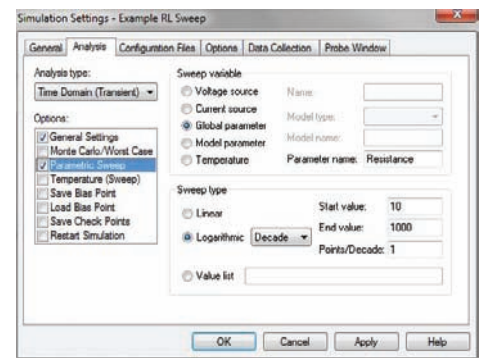
■ FIGURE 8.11 (a) Add New Column dialog box in the Property Editor for PARAM. (b) Resistor value dialog box.

Our third task consists of setting up the simulation, which includes setting transient analysis parameters as well as the values we desire for R1. Under **PSpice** we select **New Simulation Profile** (Fig. 8.12a), in which we select **Time Domain (Transient)** for **Analysis type**, 300 ns for **Run to time**, and tick the **Parametric Sweep** box under **Options**. This last action results in the dialog box shown in Fig. 8.12b, in which we select **Global parameter** for **Sweep variable** and enter *Resistance* for **Parameter name**. The final setup step required is to select **Logarithmic** under **Sweep type**, a **Start value** of 10, an **End value** of 1000, and 1 **Points/Decade**; alternatively we could list the desired resistor values using **Value list**.

After running the simulation, the notification box shown in Fig. 8.13 appears, listing the available data sets for plotting (**Resistance** = 10, 100, and 1000 in this case). A particular data set is selected by highlighting it; we select all three for this example. Upon selecting the

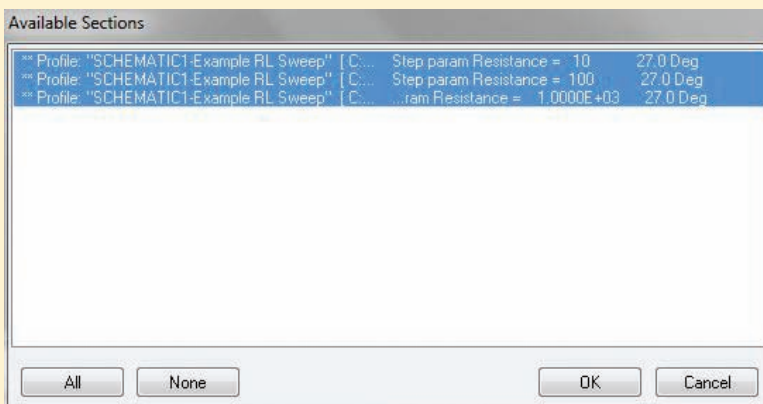


(a)



(b)

■ FIGURE 8.12 (a) Simulation dialog box. (b) Parameter sweep dialog box.



■ FIGURE 8.13 Available data sections dialog box.

(Continued on next page)

inductor current from our **Trace** variable choices in Probe, we obtain three graphs at once, as shown (after labeling by hand) in Fig. 8.14.

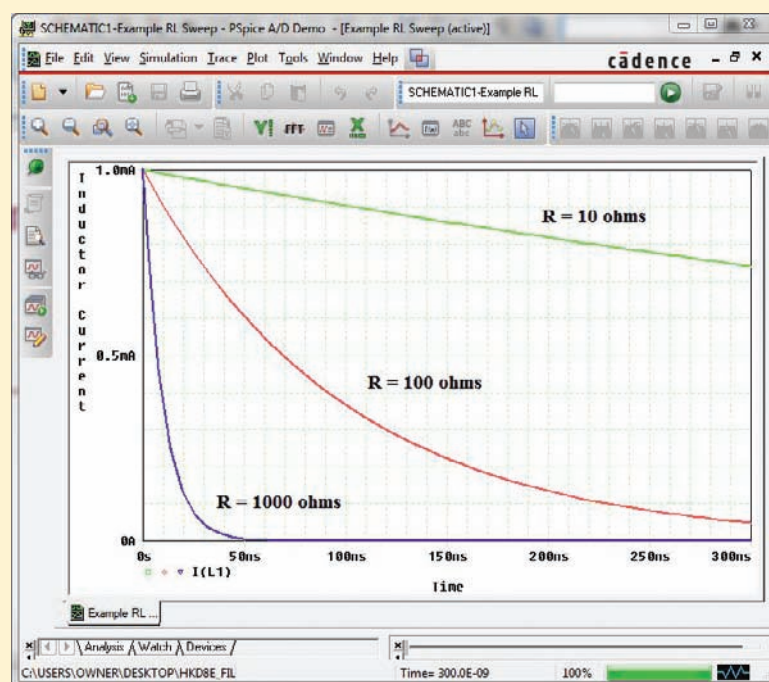


FIGURE 8.14 Probe output for the three resistances.

Why does a larger value of the time constant L/R produce a response curve that decays more slowly? Let us consider the effect of each element.

In terms of the time constant τ , the response of the series RL circuit may be written simply as

$$i(t) = I_0 e^{-t/\tau}$$

An increase in L allows a greater energy storage for the same initial current, and this larger energy requires a longer time to be dissipated in the resistor. We may also increase L/R by reducing R . In this case, the power flowing into the resistor is less for the same initial current; again, a greater time is required to dissipate the stored energy. This effect is seen clearly in our simulation result of Fig. 8.14.

8.3 THE SOURCE-FREE RC CIRCUIT

Circuits based on resistor-capacitor combinations are more common than their resistor-inductor analogs. The principal reasons for this are the smaller losses present in a physical capacitor, lower cost, better agreement between the simple mathematical model and the actual device behavior, and also smaller size and lighter weight, both of which are particularly important for integrated-circuit applications.

Let us see how closely the analysis of the parallel (or is it series?) RC circuit shown in Fig. 8.15 corresponds to that of the RL circuit. We will assume an initial stored energy in the capacitor by selecting

$$v(0) = V_0$$

The total current leaving the node at the top of the circuit diagram must be zero, so we may write

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

Division by C gives us

$$\frac{dv}{dt} + \frac{v}{RC} = 0 \quad [13]$$

Equation [13] has a familiar form; comparison with Eq. [1]

$$\frac{di}{dt} + \frac{R}{L}i = 0 \quad [1]$$

shows that the replacement of i by v and L/R by RC produces the identical equation we considered previously. It should, for the RC circuit we are now analyzing is the dual of the RL circuit we considered first. This *duality* forces $v(t)$ for the RC circuit and $i(t)$ for the RL circuit to have identical expressions if the resistance of one circuit is equal to the reciprocal of the resistance of the other circuit, and if L is numerically equal to C . Thus, the response of the RL circuit

$$i(t) = i(0)e^{-Rt/L} = I_0e^{-Rt/L}$$

enables us to immediately write

$$v(t) = v(0)e^{-t/RC} = V_0e^{-t/RC} \quad [14]$$

for the RC circuit.

Suppose instead that we had selected the current i as our variable in the RC circuit, rather than the voltage v . Applying Kirchhoff's voltage law,

$$\frac{1}{C} \int_{t_0}^t i dt' - v_0(t_0) + Ri = 0$$

we obtain an integral equation as opposed to a differential equation. However, taking the time derivative of both sides of this equation,

$$\frac{i}{C} + R \frac{di}{dt} = 0 \quad [15]$$

and replacing i with v/R , we obtain Eq. [13] again:

$$\frac{v}{RC} + \frac{dv}{dt} = 0$$

Equation [15] could have been used as our starting point, but the application of duality principles would not have been as natural.

Let us discuss the physical nature of the voltage response of the RC circuit as expressed by Eq. [14]. At $t = 0$ we obtain the correct initial condition, and as t becomes infinite, the voltage approaches zero. This latter result agrees with our thinking that if there were any voltage remaining across the capacitor, then energy would continue to flow into the resistor and be dissipated as heat. *Thus, a final voltage of zero is necessary.* The time constant of the RC

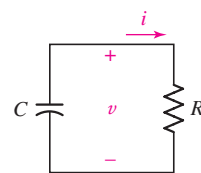


FIGURE 8.15 A parallel RC circuit for which $v(t)$ is to be determined, subject to the initial condition that $v(0) = V_0$.

circuit may be found by using the duality relationships on the expression for the time constant of the RL circuit, or it may be found by simply noting the time at which the response has dropped to 37 percent of its initial value:

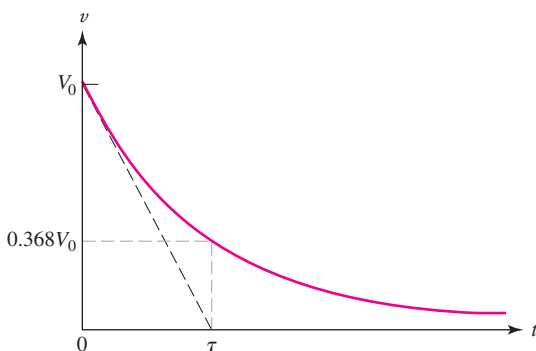
$$\frac{\tau}{RC} = 1$$

so that

$$\tau = RC \quad [16]$$



Our familiarity with the negative exponential and the significance of the time constant τ enables us to sketch the response curve readily (Fig. 8.16). Larger values of R or C provide larger time constants and slower dissipation of the stored energy. A larger resistance will dissipate a smaller power with a given voltage across it, thus requiring a greater time to convert the stored energy into heat; a larger capacitance stores a larger energy with a given voltage across it, again requiring a greater time to lose this initial energy.



■ FIGURE 8.16 The capacitor voltage $v(t)$ in the parallel RC circuit is plotted as a function of time. The initial value of $v(t)$ is V_0 .

EXAMPLE 8.3

For the circuit of Fig. 8.17a, find the voltage labeled v at $t = 200 \mu\text{s}$.

To find the requested voltage, we will need to draw and analyze two separate circuits: one corresponding to before the switch is thrown (Fig. 8.17b), and one corresponding to after the switch is thrown (Fig. 8.17c).

The sole purpose of analyzing the circuit of Fig. 8.17b is to obtain an initial capacitor voltage; we assume any transients in that circuit died out long ago, leaving a purely dc circuit. With no current through either the capacitor or the 4Ω resistor, then,

$$v(0) = 9 \text{ V} \quad [17]$$

We next turn our attention to the circuit of Fig. 8.17c, recognizing that

$$\tau = RC = (2 + 4)(10 \times 10^{-6}) = 60 \times 10^{-6} \text{ s}$$

Thus, from Eq. [14],

$$v(t) = v(0)e^{-t/RC} = v(0)e^{-t/60 \times 10^{-6}} \quad [18]$$

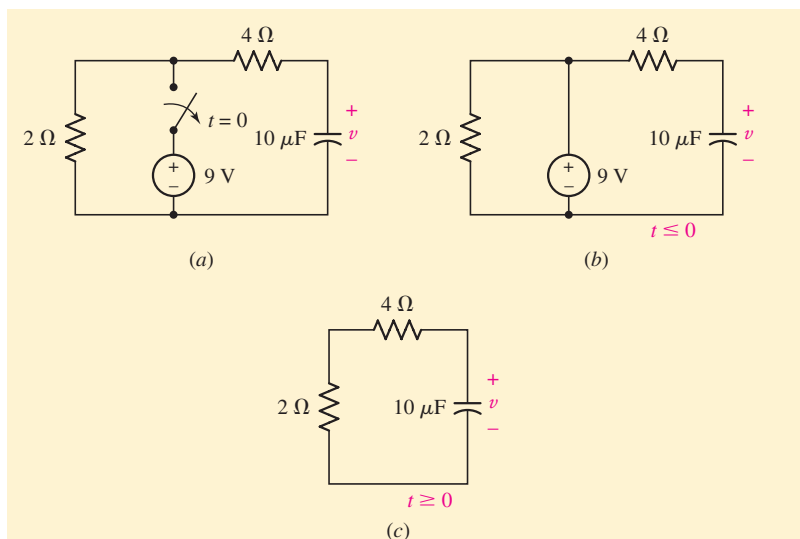


FIGURE 8.17 (a) A simple RC circuit with a switch thrown at time $t = 0$. (b) The circuit as it exists prior to $t = 0$. (c) The circuit after the switch is thrown, and the 9 V source is removed.

The capacitor voltage must be the same in both circuits at $t = 0$; no such restriction is placed on any other voltage or current. Substituting Eq. [17] into Eq. [18],

$$v(t) = 9e^{-t/60 \times 10^{-6}} \text{ V}$$

so that $v(200 \times 10^{-6}) = 321.1 \text{ mV}$ (less than 4 percent of its maximum value).

PRACTICE

8.4 Noting carefully how the circuit changes once the switch in the circuit of Fig. 8.18 is thrown, determine $v(t)$ at $t = 0$ and at $t = 160 \mu\text{s}$.

Ans: 50 V, 18.39 V.

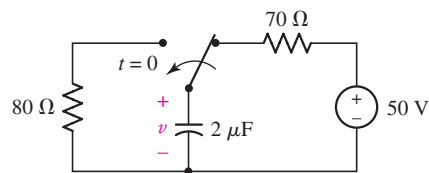


FIGURE 8.18

8.4 A MORE GENERAL PERSPECTIVE

As seen indirectly from Examples 8.2 and 8.3, regardless of how many resistors we have in the circuit, we obtain a single time constant (either $\tau = L/R$ or $\tau = RC$) when only one energy storage element is present. We can formalize this by realizing that the value needed for R is in fact the Thévenin equivalent resistance seen by our energy storage element. (Strange as it may seem, it is even possible to compute a time constant for a circuit containing dependent sources!)

General RL Circuits

As an example, consider the circuit shown in Fig. 8.19. The equivalent resistance the inductor faces is

$$R_{\text{eq}} = R_3 + R_4 + \frac{R_1 R_2}{R_1 + R_2}$$

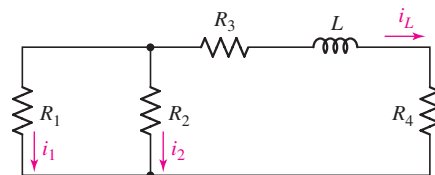


FIGURE 8.19 A source-free circuit containing one inductor and several resistors is analyzed by determining the time constant $\tau = L/R_{\text{eq}}$.

We could also write

$$\tau = \frac{L}{R_{TH}},$$

where R_{TH} is the Thévenin equivalent resistance “seen” by the inductor L .

and the time constant is therefore

$$\tau = \frac{L}{R_{eq}} \quad [19]$$

If several inductors are present in a circuit and can be combined using series and/or parallel combination, then Eq. [19] can be further generalized to

$$\tau = \frac{L_{eq}}{R_{eq}} \quad [20]$$

where L_{eq} represents the equivalent inductance.

Slicing Thinly: The Distinction Between 0^+ and 0^-

Let's return to the circuit of Fig. 8.19, and assume that some finite amount of energy is stored in the inductor at $t = 0$, so that $i_L(0) \neq 0$.

The inductor current i_L is

$$i_L = i_L(0)e^{-t/\tau}$$

and this represents what we might call the basic solution to the problem. It is quite possible that some current or voltage other than i_L is needed, such as the current i_2 in R_2 . We can always apply Kirchhoff's laws and Ohm's law to the resistive portion of the circuit without any difficulty, but current division provides the quickest answer in this circuit:

$$i_2 = -\frac{R_1}{R_1 + R_2}[i_L(0)e^{-t/\tau}]$$

It may also happen that we know the initial value of some current *other* than the inductor current. Since *the current in a resistor may change instantaneously*, we will indicate the instant after any change that might have occurred at $t = 0$ by the use of the symbol 0^+ ; in more mathematical language, $i_1(0^+)$ is the limit from the right of $i_1(t)$ as t approaches zero.¹ Thus, if we are given the initial value of i_1 as $i_1(0^+)$, then the initial value of i_2 is

$$i_2(0^+) = i_1(0^+) \frac{R_1}{R_2}$$

From these values, we obtain the necessary initial value of i_L :

$$i_L(0^+) = -[i_1(0^+) + i_2(0^+)] = -\frac{R_1 + R_2}{R_2}i_1(0^+)$$

and the expression for i_2 becomes

$$i_2 = i_1(0^+) \frac{R_1}{R_2} e^{-t/\tau}$$

Let us see if we can obtain this last expression more directly. Since the inductor current decays exponentially as $e^{-t/\tau}$, *every current throughout the circuit must follow the same functional behavior*. This is made clear by considering the inductor current as a source current that is being applied to a resistive network. Every current and voltage in the resistive network must have the same time dependence. Using these ideas, we therefore express i_2 as

$$i_2 = Ae^{-t/\tau}$$

where

$$\tau = \frac{L}{R_{eq}}$$

(1) Note that this is a notational convenience *only*. When faced with $t = 0^+$ or its companion $t = 0^-$ in an equation, we simply use the value zero. This notation allows us to clearly differentiate between the time before and after an event, such as a switch opening or closing, or a power supply being turned on or off.

Note that $i_L(0^+)$ is *always* equal to $i_L(0^-)$. This is not necessarily true for the inductor voltage or any resistor voltage or current, since they may change in zero time.



and A must be determined from a knowledge of the initial value of i_2 . Since $i_1(0^+)$ is known, the voltage across R_1 and R_2 is known, and

$$R_2 i_2(0^+) = R_1 i_1(0^+)$$

leads to

$$i_2(0^+) = i_1(0^+) \frac{R_1}{R_2}$$

Therefore,

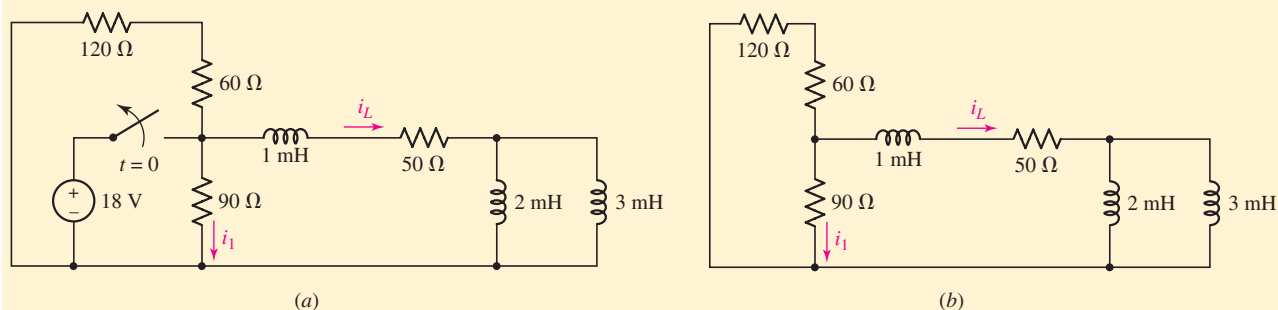
$$i_2(t) = i_1(0^+) \frac{R_1}{R_2} e^{-t/\tau}$$

A similar sequence of steps will provide a rapid solution to a large number of problems. We first recognize the time dependence of the response as an exponential decay, determine the appropriate time constant by combining resistances, write the solution with an unknown amplitude, and then determine the amplitude from a given initial condition.

This same technique can be applied to any circuit with one inductor and any number of resistors, as well as to those special circuits containing two or more inductors and also two or more resistors that may be simplified by resistance or inductance combination to one inductor and one resistor.

EXAMPLE 8.4

Determine both i_1 and i_L in the circuit shown in Fig. 8.20a for $t > 0$.



■ FIGURE 8.20 (a) A circuit with multiple resistors and inductors. (b) After $t = 0$, the circuit simplifies to an equivalent resistance of $110\ \Omega$ in series with $L_{eq} = 2.2$ mH.

After $t = 0$, when the voltage source is disconnected as shown in Fig. 8.20b, we easily calculate an equivalent inductance,

$$L_{eq} = \frac{2 \times 3}{2 + 3} + 1 = 2.2\text{ mH}$$

an equivalent resistance, in series with the equivalent inductance,

$$R_{eq} = \frac{90(60 + 120)}{90 + 180} + 50 = 110\ \Omega$$

(Continued on next page)

and the time constant,

$$\tau = \frac{L_{\text{eq}}}{R_{\text{eq}}} = \frac{2.2 \times 10^{-3}}{110} = 20 \mu\text{s}$$

Thus, the form of the natural response is $Ke^{-50,000t}$, where K is an unknown constant. Considering the circuit just prior to the switch opening ($t = 0^-$), $i_L = 18/50$ A. Since $i_L(0^+) = i_L(0^-)$, we know that $i_L = 18/50$ A or 360 mA at $t = 0^+$ and so

$$i_L = \begin{cases} 360 \text{ mA} & t < 0 \\ 360e^{-50,000t} \text{ mA} & t \geq 0 \end{cases}$$

There is no restriction on i_L changing instantaneously at $t = 0$, so its value at $t = 0^-$ (18/90 A or 200 mA) is not relevant to finding i_L for $t > 0$. Instead, we must find $i_L(0^+)$ through our knowledge of $i_L(0^+)$. Using current division,

$$i_L(0^+) = -i_L(0^+) \frac{120 + 60}{120 + 60 + 90} = -240 \text{ mA}$$

Hence,

$$i_L = \begin{cases} 200 \text{ mA} & t < 0 \\ -240e^{-50,000t} \text{ mA} & t \geq 0 \end{cases}$$

We can verify our analysis using PSpice and the switch model **Sw_tOpen**, although it should be remembered that this part is actually just two resistance values: one corresponding to before the switch opens at the specified time (the default value is 10 mΩ), and one for after the switch opens (the default value is 1 MΩ). If the equivalent resistance of the remainder of the circuit is comparable to either value, the values should be edited by double-clicking on the switch symbol in the circuit schematic. Note that there is also a switch model that closes at a specified time: **Sw_tClose**.

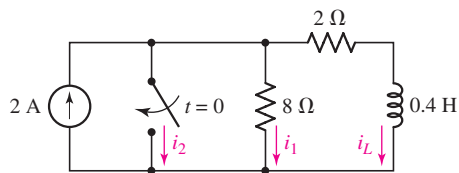


FIGURE 8.21

PRACTICE

8.5 At $t = 0.15$ s in the circuit of Fig. 8.21, find the value of (a) i_L ; (b) i_1 ; (c) i_2 .

Ans: 0.756 A; 0; 1.244 A.

We have now considered the task of finding the natural response of any circuit which can be represented by an equivalent inductor in series with an equivalent resistor. *A circuit containing several resistors and several inductors does not always possess a form which allows either the resistors or the inductors to be combined into single equivalent elements.* In such instances, there is no single negative exponential term or single time constant associated with the circuit. Rather, there will, in general, be several negative



exponential terms, the number of terms being equal to the number of inductors that remain after all possible inductor combinations have been made.

General RC Circuits

Many of the RC circuits for which we would like to find the natural response contain more than a single resistor and capacitor. Just as we did for the RL circuits, we first consider those cases in which the given circuit may be reduced to an equivalent circuit consisting of only one resistor and one capacitor.

Let us suppose first that we are faced with a circuit containing a single capacitor, but any number of resistors. It is possible to replace the two-terminal resistive network which is across the capacitor terminals with an equivalent resistor, and we may then write down the expression for the capacitor voltage immediately. In such instances, the circuit has an effective time constant given by

$$\tau = R_{\text{eq}}C$$

where R_{eq} is the equivalent resistance of the network. An alternative perspective is that R_{eq} is in fact the Thévenin equivalent resistance “seen” by the capacitor.

If the circuit has more than one capacitor, but they may be replaced somehow using series and/or parallel combinations with an equivalent capacitance C_{eq} , then the circuit has an effective time constant given by

$$\tau = RC_{\text{eq}}$$

with the general case expressed as

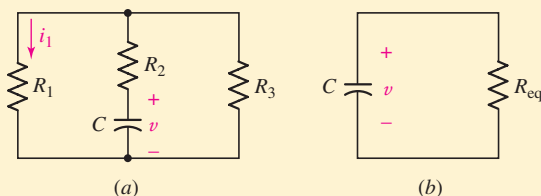
$$\tau = R_{\text{eq}}C_{\text{eq}}$$

It is worth noting, however, that parallel capacitors replaced by an equivalent capacitance would have to have identical initial conditions.



EXAMPLE 8.5

Find $v(0^+)$ and $i_1(0^+)$ for the circuit shown in Fig. 8.22a if $v(0^-) = V_0$.



■ FIGURE 8.22 (a) A given circuit containing one capacitor and several resistors. (b) The resistors have been replaced by a single equivalent resistor; the time constant is simply $\tau = R_{\text{eq}}C$.

We first simplify the circuit of Fig. 8.22a to that of Fig. 8.22b, enabling us to write

$$v = V_0 e^{-t/R_{\text{eq}}C}$$

(Continued on next page)

where

$$v(0^+) = v(0^-) = V_0 \quad \text{and} \quad R_{\text{eq}} = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

Every current and voltage in the resistive portion of the network must have the form $Ae^{-t/R_{\text{eq}}C}$, where A is the initial value of that current or voltage. Thus, the current in R_1 , for example, may be expressed as

$$i_1 = i_1(0^+)e^{-t/\tau}$$

where

$$\tau = \left(R_2 + \frac{R_1 R_3}{R_1 + R_3} \right) C$$

and $i_1(0^+)$ remains to be determined from the initial condition. Any current flowing in the circuit at $t = 0^+$ must come from the capacitor. Therefore, since v cannot change instantaneously, $v(0^+) = v(0^-) = V_0$ and

$$i_1(0^+) = \frac{V_0}{R_2 + R_1 R_3 / (R_1 + R_3)} \frac{R_3}{R_1 + R_3}$$

PRACTICE

8.6 Find values of v_C and v_o in the circuit of Fig. 8.23 at t equal to (a) 0^- ; (b) 0^+ ; (c) 1.3 ms.

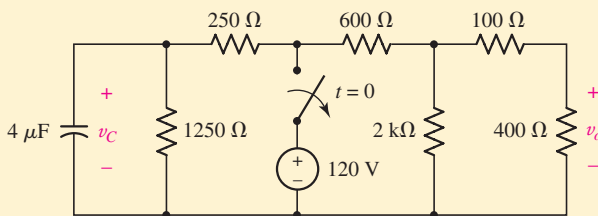


FIGURE 8.23

Ans: 100 V, 38.4 V; 100 V, 25.6 V; 59.5 V, 15.22 V.

Our method can be applied to circuits with one energy storage element and one or more dependent sources as well. In such instances, we may write an appropriate KCL or KVL equation along with any necessary supporting equations, distill this down into a single differential equation, and extract the characteristic equation to find the time constant. Alternatively, we may begin by finding the Thévenin equivalent resistance of the network connected to the capacitor or inductor, and use this in calculating the appropriate RL or RC time constant—unless the dependent source is controlled by a voltage or current associated with the energy storage element, in which case the Thévenin approach cannot be used.

EXAMPLE 8.6

For the circuit of Fig. 8.24a, find the voltage labeled v_C for $t > 0$ if $v_C(0^-) = 2$ V.

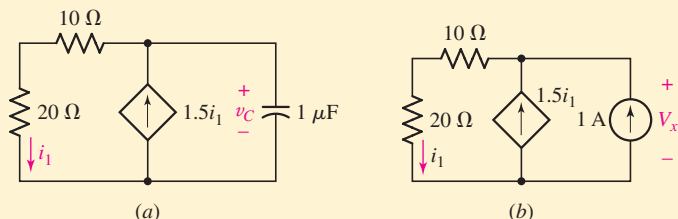


FIGURE 8.24 (a) A simple RC circuit containing a dependent source not controlled by a capacitor voltage or current. (b) Circuit for finding the Thévenin equivalent of the network connected to the capacitor.

The dependent source is not controlled by a capacitor voltage or current, so we can start by finding the Thévenin equivalent of the network to the left of the capacitor. Connecting a 1 A test source as in Fig. 8.24b,

$$V_x = (1 + 1.5i_1)(30)$$

where

$$i_1 = \left(\frac{1}{20}\right) \frac{20}{10 + 20} V_x = \frac{V_x}{30}$$

Performing a little algebra, we find that $V_x = -60$ V, so the network has a Thévenin equivalent resistance of -60 Ω (unusual, but not impossible when dealing with a dependent source). Our circuit therefore has a *negative* time constant

$$\tau = -60(1 \times 10^{-6}) = -60 \mu s$$

The capacitor voltage is therefore

$$v_C(t) = Ae^{t/60 \times 10^{-6}} \quad \text{V}$$

where $A = v_C(0^+) = v_C(0^-) = 2$ V. Thus,

$$v_C(t) = 2e^{t/60 \times 10^{-6}} \quad \text{V} \quad [21]$$

which, interestingly enough is unstable: it grows exponentially with time. This cannot continue indefinitely; one or more elements in the circuit will eventually fail.

Alternatively, we could write a simple KCL equation for the top node of Fig. 8.24a

$$v_C = 30 \left(1.5i_1 - 10^{-6} \frac{dv_C}{dt} \right) \quad [22]$$

where

$$i_1 = \frac{v_C}{30} \quad [23]$$

(Continued on next page)

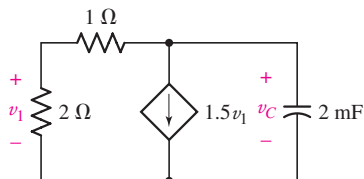


FIGURE 8.25 Circuit for Practice Problem 8.7.

Substituting Eq. [23] into Eq. [22] and performing some algebra, we obtain

$$\frac{dv_C}{dt} - \frac{1}{60 \times 10^{-6}} v_C = 0$$

which has the characteristic equation

$$s - \frac{1}{60 \times 10^{-6}} = 0$$

Thus,

$$s = \frac{1}{60 \times 10^{-6}}$$

and so

$$v_C(t) = Ae^{t/60 \times 10^{-6}} \quad \text{V}$$

as we found before. Substitution of $A = v_C(0^+) = 2$ results in Eq. [21], our expression for the capacitor voltage for $t > 0$.

PRACTICE

8.7 (a) Regarding the circuit of Fig. 8.25, determine the voltage $v_C(t)$ for $t > 0$ if $v_C(0^-) = 11$ V. (b) Is the circuit “stable”?

Ans: (a) $v_C(t) = 11e^{-2 \times 10^3 t/3}$ V, $t > 0$. (b) Yes; it decays (exponentially) rather than grows with time.



Some circuits containing a number of both resistors and capacitors may be replaced by an equivalent circuit containing only one resistor and one capacitor; it is necessary that the original circuit be one which can be broken into two parts, one containing all resistors and the other containing all capacitors, such that the two parts are connected by only two ideal conductors. Otherwise, multiple time constants and multiple exponential terms will be required to describe the behavior of the circuit (one time constant for each energy storage element remaining in the circuit after it is reduced as much as possible).

As a parting comment, we should be wary of certain situations involving only ideal elements which are suddenly connected together. For example, we may imagine connecting two ideal capacitors in series having unequal voltages prior to $t = 0$. This poses a problem using our mathematical model of an ideal capacitor; however, real capacitors have resistances associated with them through which energy can be dissipated.

8.5 THE UNIT-STEP FUNCTION

We have been studying the response of RL and RC circuits when no sources or forcing functions were present. We termed this response the *natural response*, because its form depends only on the nature of the circuit. The reason that any response at all is obtained arises from the presence of initial

energy storage within the inductive or capacitive elements in the circuit. In some cases we were confronted with circuits containing sources and switches; we were informed that certain switching operations were performed at $t = 0$ in order to remove all the sources from the circuit, while leaving known amounts of energy stored here and there. In other words, we have been solving problems in which energy sources are suddenly *removed* from the circuit; now we must consider that type of response which results when energy sources are suddenly *applied* to a circuit.

We will focus on the response which occurs when the energy sources suddenly applied are dc sources. Since every electrical device is intended to be energized at least once, and since most devices are turned on and off many times in the course of their lifetimes, our study applies to many practical cases. Even though we are now restricting ourselves to dc sources, there are still many cases in which these simpler examples correspond to the operation of physical devices. For example, the first circuit we will analyze could represent the buildup of the current when a dc motor is started. The generation and use of the rectangular voltage pulses needed to represent a number or a command in a microprocessor provide many examples in the field of electronic or transistor circuitry. Similar circuits are found in the synchronization and sweep circuits of television receivers, in communication systems using pulse modulation, and in radar systems, to name but a few examples.

We have been speaking of the “sudden application” of an energy source, and by this phrase we imply its application in zero time.² The operation of a switch in series with a battery is thus equivalent to a forcing function which is zero up to the instant that the switch is closed and is equal to the battery voltage thereafter. The forcing function has a break, or discontinuity, at the instant the switch is closed. Certain special forcing functions which are discontinuous or have discontinuous derivatives are called **singularity functions**, the two most important of these singularity functions being the **unit-step function** and the **unit-impulse function**.

We define the unit-step forcing function as a function of time which is zero for all values of its argument less than zero and which is unity for all positive values of its argument. If we let $(t - t_0)$ be the argument and represent the unit-step function by u , then $u(t - t_0)$ must be zero for all values of t less than t_0 , and it must be unity for all values of t greater than t_0 . At $t = t_0$, $u(t - t_0)$ changes *abruptly* from 0 to 1. Its value at $t = t_0$ is not defined, but its value is known for all instants of time that are arbitrarily close to $t = t_0$. We often indicate this by writing $u(t_0^-) = 0$ and $u(t_0^+) = 1$. The concise mathematical definition of the unit-step forcing function is

$$u(t - t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t > t_0 \end{cases}$$

and the function is shown graphically in Fig. 8.26. Note that a vertical line of unit length is shown at $t = t_0$. Although this “riser” is not strictly a part of the definition of the unit step, it is usually shown in each drawing.

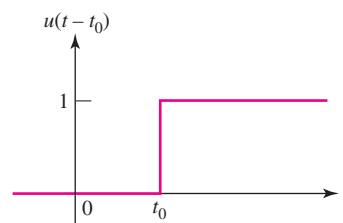
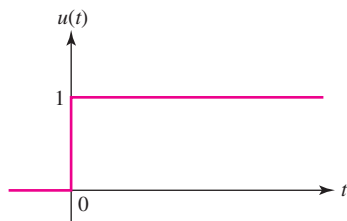


FIGURE 8.26 The unit-step forcing function, $u(t - t_0)$.

(2) Of course, this is not physically possible. However, if the time scale over which such an event occurs is very short compared to all other relevant time scales that describe the operation of a circuit, this is approximately true, and mathematically convenient.



■ FIGURE 8.27 The unit-step forcing function $u(t)$ is shown as a function of t .

We also note that the unit step need not be a time function. For example, $u(x - x_0)$ could be used to denote a unit-step function where x might be a distance in meters, for example, or a frequency.

Very often in circuit analysis a discontinuity or a switching action takes place at an instant that is defined as $t = 0$. In that case $t_0 = 0$, and we then represent the corresponding unit-step forcing function by $u(t - 0)$, or more simply $u(t)$. This is shown in Fig. 8.27. Thus

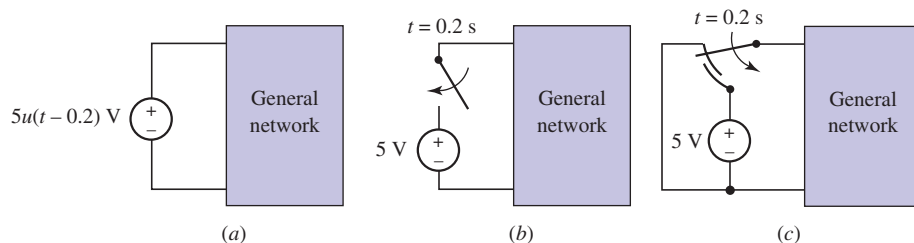
$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

The unit-step forcing function is in itself dimensionless. If we wish it to represent a voltage, it is necessary to multiply $u(t - t_0)$ by some constant voltage, such as 5 V. Thus, $v(t) = 5u(t - 0.2)$ V is an ideal voltage source which is zero before $t = 0.2$ s and a constant 5 V after $t = 0.2$ s. This forcing function is shown connected to a general network in Fig. 8.28a.

Physical Sources and the Unit-Step Function

Perhaps we should ask what physical source is the equivalent of this discontinuous forcing function. By equivalent, we mean simply that the voltage-current characteristics of the two networks are identical. For the step-voltage source of Fig. 8.28a, the voltage-current characteristic is simple: the voltage is zero prior to $t = 0.2$ s, it is 5 V after $t = 0.2$ s, and the current may be any (finite) value in either time interval. Our first thoughts might produce the attempt at an equivalent shown in Fig. 8.28b, a 5 V dc source in series with a switch which closes at $t = 0.2$ s. This network is not equivalent for $t < 0.2$ s, however, because the voltage across the battery and switch is completely unspecified in this time interval. The “equivalent” source is an open circuit, and the voltage across it *may be anything*. After $t = 0.2$ s, the networks are equivalent, and if this is the only time interval in which we are interested, and if the initial currents which flow from the two networks are identical at $t = 0.2$ s, then Fig. 8.28b becomes a useful equivalent of Fig. 8.28a.

In order to obtain an exact equivalent for the voltage-step forcing function, we may provide a single-pole double-throw switch. Before $t = 0.2$ s, the switch serves to ensure zero voltage across the input terminals of the general network. After $t = 0.2$ s, the switch is thrown to provide a constant input voltage of 5 V. At $t = 0.2$ s, the voltage is indeterminate (as is the step forcing function), and the battery is momentarily short-circuited (it is



■ FIGURE 8.28 (a) A voltage-step forcing function is shown as the source driving a general network. (b) A simple circuit which, although not the exact equivalent of part (a), may be used as its equivalent in many cases. (c) An exact equivalent of part (a).

fortunate that we are dealing with mathematical models!). This exact equivalent of Fig. 8.28a is shown in Fig. 8.28c.

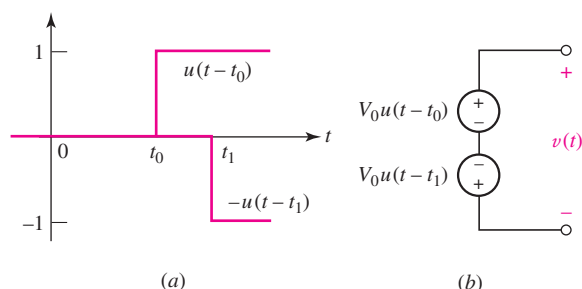
Figure 8.29a shows a current-step forcing function driving a general network. If we attempt to replace this circuit by a dc source in parallel with a switch (which opens at $t = t_0$), we must realize that the circuits are equivalent after $t = t_0$ but that the responses after $t = t_0$ are alike only if the initial conditions are the same. The circuit in Fig. 8.29b implies no voltage exists across the current source terminals for $t < t_0$. This is not the case for the circuit of Fig. 8.29a. However, we may often use the circuits of Fig. 8.29a and b interchangeably. The exact equivalent of Fig. 8.29a is the dual of the circuit of Fig. 8.28c; the exact equivalent of Fig. 8.29b cannot be constructed with current- and voltage-step forcing functions alone.³

The Rectangular Pulse Function

Some very useful forcing functions may be obtained by manipulating the unit-step forcing function. Let us define a rectangular voltage pulse by the following conditions:

$$v(t) = \begin{cases} 0 & t < t_0 \\ V_0 & t_0 < t < t_1 \\ 0 & t > t_1 \end{cases}$$

The pulse is drawn in Fig. 8.30. Can this pulse be represented in terms of the unit-step forcing function? Let us consider the difference of the two unit steps, $u(t - t_0) - u(t - t_1)$. The two step functions are shown in Fig. 8.31a, and their difference is a rectangular pulse. The source $V_0 u(t - t_0) - V_0 u(t - t_1)$ which provides us with the desired voltage is indicated in Fig. 8.31b.



■ FIGURE 8.31 (a) The unit steps $u(t - t_0)$ and $-u(t - t_1)$. (b) A source which yields the rectangular voltage pulse of Fig. 8.30.

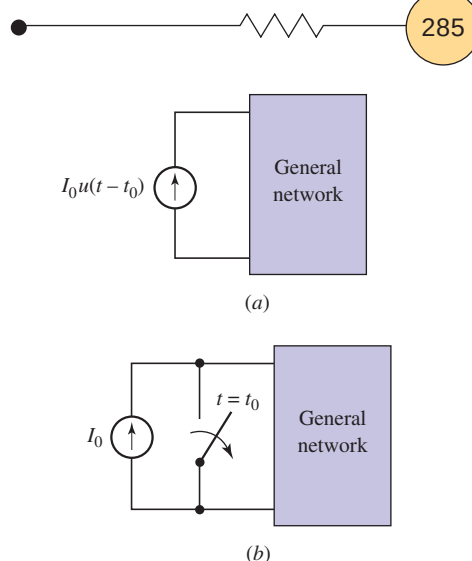
If we have a sinusoidal voltage source $V_m \sin \omega t$ which is suddenly connected to a network at $t = t_0$, then an appropriate voltage forcing function would be $v(t) = V_m u(t - t_0) \sin \omega t$. If we wish to represent one burst of energy from the transmitter for a radio-controlled car operating at 47 MHz (295 Mrad/s), we may turn the sinusoidal source off 70 ns later by a second unit-step forcing function.⁴ The voltage pulse is thus

$$v(t) = V_m [u(t - t_0) - u(t - t_0 - 7 \times 10^{-8})] \sin(295 \times 10^6 t)$$

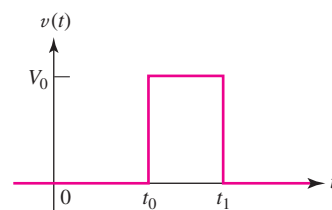
This forcing function is sketched in Fig. 8.32.

(3) The equivalent can be drawn if the current through the switch prior to $t = t_0$ is known.

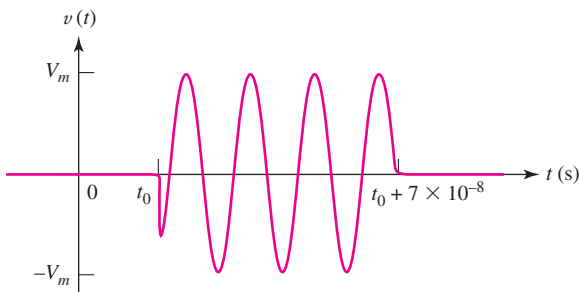
(4) Apparently, we're pretty good at the controls of this car. A reaction time of 70 ns?



■ FIGURE 8.29 (a) A current-step forcing function is applied to a general network. (b) A simple circuit which, although not the exact equivalent of part (a), may be used as its equivalent in many cases.



■ FIGURE 8.30 A useful forcing function, the rectangular voltage pulse.

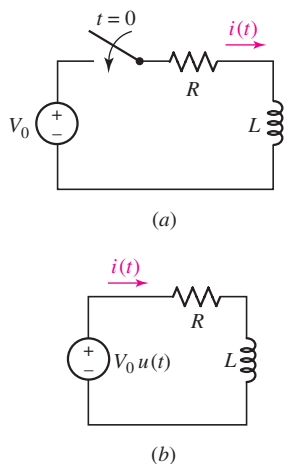


■ FIGURE 8.32 A 47 MHz radio-frequency pulse, described by $v(t) = V_m[u(t - t_0) - u(t - t_0 - 7 \times 10^{-8})] \sin(259 \times 10^6 t)$.

PRACTICE

8.8 Evaluate each of the following at $t = 0.8$: (a) $3u(t) - 2u(-t) + 0.8u(1 - t)$; (b) $[4u(t)]u(-t)$; (c) $2u(t) \sin \pi t$.

Ans: 3.8; 0; 1.176.



■ FIGURE 8.33 (a) The given circuit. (b) An equivalent circuit, possessing the same response $i(t)$ for all time.

8.6 DRIVEN RL CIRCUITS

We are now ready to subject a simple network to the sudden application of a dc source. The circuit consists of a battery whose voltage is V_0 in series with a switch, a resistor R , and an inductor L . The switch is closed at $t = 0$, as indicated on the circuit diagram of Fig. 8.33a. It is evident that the current $i(t)$ is zero before $t = 0$, and we are therefore able to replace the battery and switch by a voltage-step forcing function $V_0 u(t)$, which also produces no response prior to $t = 0$. After $t = 0$, the two circuits are clearly identical. Hence, we seek the current $i(t)$ either in the given circuit of Fig. 8.33a or in the equivalent circuit of Fig. 8.33b.

We will find $i(t)$ at this time by writing the appropriate circuit equation and then solving it by separation of the variables and integration. After we obtain the answer and investigate the two parts of which it is composed, we will see that there is physical significance to each of these two terms. With a more intuitive understanding of how each term originates, we will be able to produce more rapid and more meaningful solutions to every problem involving the sudden application of any source.

Applying Kirchhoff's voltage law to the circuit of Fig. 8.33b, we have

$$Ri + L \frac{di}{dt} = V_0 u(t)$$

Since the unit-step forcing function is discontinuous at $t = 0$, we will first consider the solution for $t < 0$ and then for $t > 0$. The application of zero voltage since $t = -\infty$ forces a zero response, so that

$$i(t) = 0 \quad t < 0$$

For positive time, however, $u(t)$ is unity and we must solve the equation

$$Ri + L \frac{di}{dt} = V_0 \quad t > 0$$

The variables may be separated in several simple algebraic steps, yielding

$$\frac{L di}{V_0 - Ri} = dt$$

and each side may be integrated directly:

$$-\frac{L}{R} \ln(V_0 - Ri) = t + k$$

In order to evaluate k , an initial condition must be invoked. Prior to $t = 0$, $i(t)$ is zero, and thus $i(0^-) = 0$. Since the current in an inductor cannot change by a finite amount in zero time without being associated with an infinite voltage, we thus have $i(0^+) = 0$. Setting $i = 0$ at $t = 0$, we obtain

$$-\frac{L}{R} \ln V_0 = k$$

and, hence,

$$-\frac{L}{R} [\ln(V_0 - Ri) - \ln V_0] = t$$

Rearranging,

$$\frac{V_0 - Ri}{V_0} = e^{-Rt/L}$$

or

$$i = \frac{V_0}{R} - \frac{V_0}{R} e^{-Rt/L} \quad t > 0 \quad [24]$$

Thus, an expression for the response valid for all t would be

$$i = \left(\frac{V_0}{R} - \frac{V_0}{R} e^{-Rt/L} \right) u(t) \quad [25]$$

A More Direct Procedure

This is the desired solution, but it has not been obtained in the simplest manner. In order to establish a more direct procedure, let us try to interpret the two terms appearing in Eq. [25]. The exponential term has the functional form of the natural response of the RL circuit; it is a negative exponential, it approaches zero as time increases, and it is characterized by the time constant L/R . The functional form of this part of the response is thus identical with that which is obtained in the source-free circuit. However, the amplitude of this exponential term depends on the source voltage V_0 . We might generalize, then, that the response will be the sum of two terms, where one term has a functional form identical to that of the source-free response, but has an amplitude that depends on the forcing function. But what of the other term?

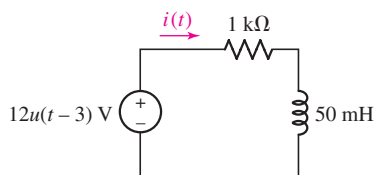
Equation [25] also contains a constant term, V_0/R . Why is it present? The answer is simple: the natural response approaches zero as the energy is gradually dissipated, but the total response must not approach zero. Eventually the circuit behaves as a resistor and an inductor in series with a battery. Since the inductor looks like a short circuit to dc, the only current now



flowing is V_0/R . This current is a part of the response that is directly attributable to the forcing function, and we call it the *forced response*. It is the response that is present a long time after the switch is closed.

The complete response is composed of two parts, the *natural response* and the *forced response*. The natural response is a characteristic of the circuit and not of the sources. Its form may be found by considering the source-free circuit, and it has an amplitude that depends on both the initial amplitude of the source and the initial energy storage. The forced response has the characteristics of the forcing function; it is found by pretending that all switches were thrown a long time ago. Since we are presently concerned only with switches and dc sources, the forced response is merely the solution of a simple dc circuit problem.

EXAMPLE 8.7



■ FIGURE 8.34 A simple RL circuit driven by a voltage-step forcing function.

For the circuit of Fig. 8.34, find $i(t)$ for $t = \infty$, 3^- , 3^+ , and $100 \mu\text{s}$ after the source changes value.

Long after any transients have died out ($t \rightarrow \infty$), the circuit is a simple dc circuit driven by a 12 V voltage source. The inductor appears as a short circuit, so

$$i(\infty) = \frac{12}{1000} = 12 \text{ mA}$$

What is meant by $i(3^-)$? This is simply a notational convenience to indicate the instant before the voltage source changes value. For $t < 3$, $u(t-3) = 0$. Thus, $i(3^-) = 0$ as well.

At $t = 3^+$, the forcing function $12u(t-3) = 12 \text{ V}$. However, since the inductor current cannot change in zero time, $i(3^+) = i(3^-) = 0$.

The most straightforward approach to analyzing the circuit for $t > 3 \text{ s}$ is to rewrite Eq. [25] as

$$i(t') = \left(\frac{V_0}{R} - \frac{V_0}{R} e^{-Rt'/L} \right) u(t')$$

and note that this equation applies to our circuit as well if we shift the time axis such that

$$t' = t - 3$$

Therefore, with $V_0/R = 12 \text{ mA}$ and $R/L = 20,000 \text{ s}^{-1}$,

$$i(t-3) = (12 - 12e^{-20,000(t-3)}) u(t-3) \quad \text{mA} \quad [26]$$

which can be written more simply as

$$i(t) = (12 - 12e^{-20,000(t-3)}) u(t-3) \quad \text{mA} \quad [27]$$

since the unit-step function forces a zero value for $t < 3$, as required.

Substituting $t = 3.0001 \text{ s}$ into Eq. [26] or [27], we find that $i = 10.38 \text{ mA}$ at a time $100 \mu\text{s}$ after the source changes value.

PRACTICE

8.9 The voltage source $60 - 40u(t)$ V is in series with a $10\ \Omega$ resistor and a 50 mH inductor. Find the magnitudes of the inductor current and voltage at t equal to (a) 0^- ; (b) 0^+ ; (c) ∞ ; (d) 3 ms.

Ans: 6 A, 0 V; 6 A, 40 V; 2 A, 0 V; 4.20 A, 22.0 V.

Developing an Intuitive Understanding

The reason for the two responses, forced and natural, may be seen from physical arguments. We know that our circuit will eventually assume the forced response. However, at the instant the switches are thrown, the initial inductor currents (or, in RC circuits, the voltages across the capacitors) will have values that depend only on the energy stored in these elements. These currents or voltages cannot be expected to be the same as the currents and voltages demanded by the forced response. Hence, there must be a transient period during which the currents and voltages change from their given initial values to their required final values. The portion of the response that provides the transition from initial to final values is the natural response (often called the *transient* response, as we found earlier). If we describe the response of the simple source-free RL circuit in these terms, then we should say that the forced response is zero and that the natural response serves to connect the initial response dictated by the stored energy with the zero value of the forced response.

This description is appropriate only for those circuits in which the natural response eventually dies out. This always occurs in physical circuits where some resistance is associated with every element, but there are a number of “pathologic” circuits in which the natural response is nonvanishing as time becomes infinite. Those circuits in which trapped currents circulate around inductive loops, or voltages are trapped in series strings of capacitors, are examples.

8.7 NATURAL AND FORCED RESPONSE

There is also an excellent mathematical reason for considering the complete response to be composed of two parts—the forced response and the natural response. The reason is based on the fact that the solution of any linear differential equation may be expressed as the sum of two parts: the *complementary solution* (natural response) and the *particular solution* (forced response). Without delving into the general theory of differential equations, let us consider a general equation of the type met in the previous section:

$$\frac{di}{dt} + Pi = Q$$

or

$$di + Pi\,dt = Q\,dt \quad [28]$$

We may identify Q as a forcing function and express it as $Q(t)$ to emphasize its general time dependence. Let us simplify the discussion by

assuming that P is a positive constant. Later, we will also assume that Q is constant, thus restricting ourselves to dc forcing functions.

In any standard text on elementary differential equations, it is shown that if both sides of Eq. [28] are multiplied by a suitable “integrating factor,” then each side becomes an exact differential that can be integrated directly to obtain the solution. We are not separating the variables, but merely arranging them in such a way that integration is possible. For this equation, the integrating factor is $e^{\int P dt}$ or simply e^{Pt} , since P is a constant. We multiply each side of the equation by this integrating factor and obtain

$$e^{Pt} di + iPe^{Pt} dt = Qe^{Pt} dt \quad [29]$$

The form of the left side may be simplified by recognizing it as the exact differential of ie^{Pt} :

$$d(ie^{Pt}) = e^{Pt} di + iPe^{Pt} dt$$

so that Eq. [29] becomes

$$d(ie^{Pt}) = Qe^{Pt} dt$$

Integrating each side,

$$ie^{Pt} = \int Qe^{Pt} dt + A$$

where A is a constant of integration. Multiplication by e^{-Pt} produces the solution for $i(t)$,

$$i = e^{-Pt} \int Qe^{Pt} dt + Ae^{-Pt} \quad [30]$$

If our forcing function $Q(t)$ is known, then we can obtain the functional form of $i(t)$ by evaluating the integral. We will not evaluate such an integral for each problem, however; instead, we are interested in using Eq. [30] to draw several very general conclusions.

The Natural Response

We note first that, for a source-free circuit, Q must be zero, and the solution is the natural response

$$i_n = Ae^{-Pt} \quad [31]$$

We will find that the constant P is never negative for a circuit with only resistors, inductors, and capacitors; its value depends only on the passive circuit elements⁵ and their interconnection in the circuit. *The natural response therefore approaches zero as time increases without limit.* This must be the case for the simple RL circuit, because the initial energy is gradually dissipated in the resistor, leaving the circuit in the form of heat. There are also idealized circuits in which P is zero; in these circuits the natural response does not die out.

We therefore find that one of the two terms making up the complete response has the form of the natural response; it has an amplitude which will

(5) If the circuit contains a dependent source or a negative resistance, P may be negative.

depend on (but *not* always be equal to) the initial value of the complete response and thus on the initial value of the forcing function also.

The Forced Response

We next observe that the first term of Eq. [30] depends on the functional form of $Q(t)$, the forcing function. Whenever we have a circuit in which the natural response dies out as t becomes infinite, this first term must completely describe the form of the response after the natural response has disappeared. This term is typically called the forced response; it is also called the *steady-state response*, the *particular solution*, or the *particular integral*.

For the present, we have elected to consider only those problems involving the sudden application of dc sources, and $Q(t)$ will therefore be a constant for all values of time. If we wish, we can now evaluate the integral in Eq. [30], obtaining the forced response

$$i_f = \frac{Q}{P} \quad [32]$$

and the complete response

$$i(t) = \frac{Q}{P} + Ae^{-Pt} \quad [33]$$

For the RL series circuit, Q/P is the constant current V_0/R and $1/P$ is the time constant τ . We should see that the forced response might have been obtained without evaluating the integral, because it must be the complete response at infinite time; it is merely the source voltage divided by the series resistance. The forced response is thus obtained by inspection of the final circuit.

Determination of the Complete Response

Let us use the simple RL series circuit to illustrate how to determine the complete response by the addition of the natural and forced responses. The circuit shown in Fig. 8.35 was analyzed earlier, but by a longer method. The desired response is the current $i(t)$, and we first express this current as the sum of the natural and the forced current,

$$i = i_n + i_f$$

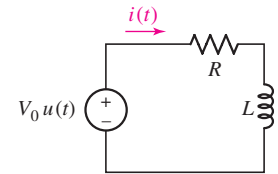
The functional form of the natural response must be the same as that obtained without any sources. We therefore replace the step-voltage source by a short circuit and recognize the old RL series loop. Thus,

$$i_n = Ae^{-Rt/L}$$

where the amplitude A is yet to be determined; since the initial condition applies to the *complete* response, we cannot simply assume $A = i(0)$.

We next consider the forced response. In this particular problem the forced response must be constant, because the source is a constant V_0 for all positive values of time. After the natural response has died out, there can be no voltage across the inductor; hence, a voltage V_0 appears across R , and the forced response is simply

$$i_f = \frac{V_0}{R}$$



■ FIGURE 8.35 A series RL circuit that is used to illustrate the method by which the complete response is obtained as the sum of the natural and forced responses.



Note that the forced response is determined completely; there is no unknown amplitude. We next combine the two responses to obtain

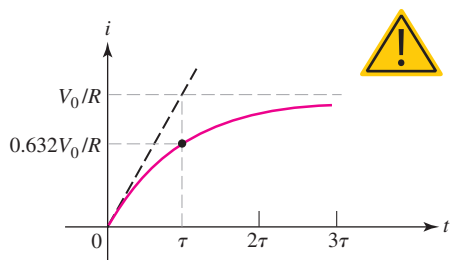
$$i = Ae^{-Rt/L} + \frac{V_0}{R}$$

and apply the initial condition to evaluate A . The current is zero prior to $t = 0$, and it cannot change value instantaneously since it is the current flowing through an inductor. Thus, the current is zero immediately after $t = 0$, and

$$0 = A + \frac{V_0}{R}$$

so

$$i = \frac{V_0}{R}(1 - e^{-Rt/L}) \quad [34]$$



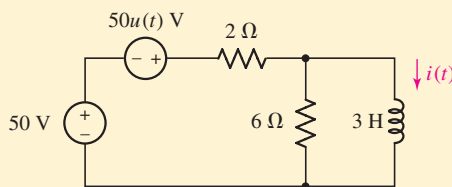
■ FIGURE 8.36 The current flowing through the inductor of Fig. 8.35 is shown graphically. A line extending the initial slope meets the constant forced response at $t = \tau$.

Note carefully that A is not the initial value of i , since $A = -V_0/R$, while $i(0) = 0$. In considering source-free circuits, we found that A was the initial value of the response. When forcing functions are present, however, we must first find the initial value of the response and then substitute this in the equation for the complete response to find A .

This response is plotted in Fig. 8.36, and we can see the manner in which the current builds up from its initial value of zero to its final value of V_0/R . The transition is effectively accomplished in a time 3τ . If our circuit represents the field coil of a large dc motor, we might have $L = 10$ H and $R = 20$ Ω , obtaining $\tau = 0.5$ s. The field current is thus established in about 1.5 s. In one time constant, the current has attained 63.2 percent of its final value.

EXAMPLE 8.8

Determine $i(t)$ for all values of time in the circuit of Fig. 8.37.



■ FIGURE 8.37 The circuit of Example 8.8.

The circuit contains a dc voltage source as well as a step-voltage source. We might choose to replace everything to the left of the inductor by the Thévenin equivalent, but instead let us merely recognize the form of that equivalent as a resistor in series with some voltage source. The circuit contains only one energy storage element, the inductor. We first note that

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{3}{1.5} = 2 \text{ s}$$

and recall that

$$i = i_f + i_n$$

The natural response is therefore a negative exponential as before:

$$i_n = Ke^{-t/2} \quad \text{A} \quad t > 0$$

Since the forcing function is a dc source, the forced response will be a constant current. The inductor acts like a short circuit to dc, so that

$$i_f = \frac{100}{2} = 50 \text{ A}$$

Thus,

$$i = 50 + Ke^{-0.5t} \quad \text{A} \quad t > 0$$

In order to evaluate K , we must establish the initial value of the inductor current. Prior to $t = 0$, this current is 25 A, and it cannot change instantaneously. Thus,

$$25 = 50 + K$$

or

$$K = -25$$

Hence,

$$i = 50 - 25e^{-0.5t} \quad \text{A} \quad t > 0$$

We complete the solution by also stating

$$i = 25 \text{ A} \quad t < 0$$

or by writing a single expression valid for all t ,

$$i = 25 + 25(1 - e^{-0.5t})u(t) \quad \text{A}$$

The complete response is sketched in Fig. 8.38. Note how the natural response serves to connect the response for $t < 0$ with the constant forced response.

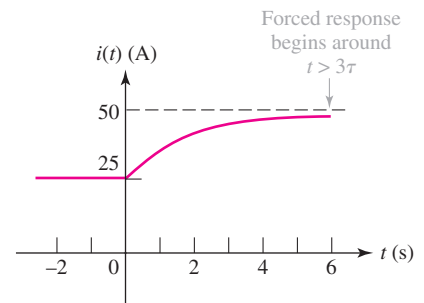


FIGURE 8.38 The response $i(t)$ of the circuit shown in Fig. 8.37 is sketched for values of times less and greater than zero.

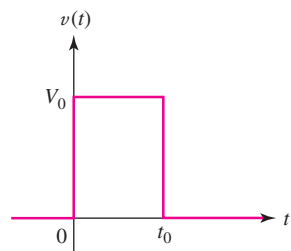
PRACTICE

8.10 A voltage source, $v_s = 20u(t)$ V, is in series with a $200 \, \Omega$ resistor and a 4 H inductor. Find the magnitude of the inductor current at t equal to (a) 0^- ; (b) 0^+ ; (c) 8 ms; (d) 15 ms.

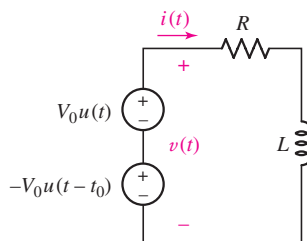
Ans: 0; 0; 33.0 mA; 52.8 mA.

As a final example of this method by which the complete response of any circuit subjected to a transient may be written down *almost by inspection*, let us again consider the simple RL series circuit, but subjected to a voltage pulse.

EXAMPLE 8.9

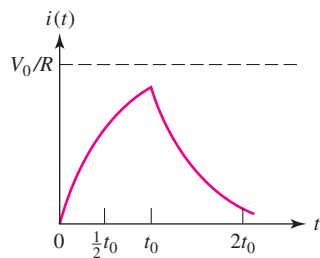


(a)

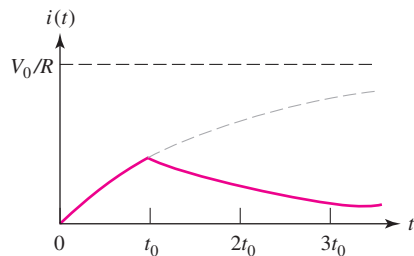


(b)

FIGURE 8.39 (a) A rectangular voltage pulse which is to be used as the forcing function in a simple series RL circuit. (b) The series RL circuit, showing the representation of the forcing function by the series combination of two independent voltage-step sources. The current $i(t)$ is desired.



(a)



(b)

FIGURE 8.40 Two possible response curves are shown for the circuit of Fig. 8.39b. (a) τ is selected as $t_0/2$. (b) τ is selected as $2t_0$.

Find the current response in a simple series RL circuit when the forcing function is a rectangular voltage pulse of amplitude V_0 and duration t_0 .

We represent the forcing function as the sum of two step-voltage sources $V_0 u(t)$ and $-V_0 u(t - t_0)$, as indicated in Fig. 8.39a and b, and we plan to obtain the response by using superposition. Let $i_1(t)$ designate that part of $i(t)$ which is due to the upper source $V_0 u(t)$ acting alone, and let $i_2(t)$ represent that part due to $-V_0 u(t - t_0)$ acting alone. Then,

$$i(t) = i_1(t) + i_2(t)$$

Our object is now to write each of the partial responses i_1 and i_2 as the sum of a natural and a forced response. The response $i_1(t)$ is familiar; this problem was solved in Eq. [34]:

$$i_1(t) = \frac{V_0}{R}(1 - e^{-Rt/L}) \quad t > 0$$

Note that this solution is only valid for $t > 0$ as indicated; $i_1 = 0$ for $t < 0$.

We now turn our attention to the other source and its response $i_2(t)$. Only the polarity of the source and the time of its application are different. There is no need therefore to determine the form of the natural response and the forced response; the solution for $i_1(t)$ enables us to write

$$i_2(t) = -\frac{V_0}{R}[1 - e^{-R(t-t_0)/L}] \quad t > t_0$$

where the applicable range of t , $t > t_0$, must again be indicated; and $i_2 = 0$ for $t < t_0$.

We now add the two solutions, but do so carefully, since each is valid over a different interval of time. Thus,

$$i(t) = 0 \quad t < 0 \quad [35]$$

$$i(t) = \frac{V_0}{R}(1 - e^{-Rt/L}) \quad 0 < t < t_0 \quad [36]$$

and

$$i(t) = \frac{V_0}{R}(1 - e^{-Rt/L}) - \frac{V_0}{R}(1 - e^{-R(t-t_0)/L}) \quad t > t_0$$

or more compactly,

$$i(t) = \frac{V_0}{R}e^{-Rt/L}(e^{Rt_0/L} - 1) \quad t > t_0 \quad [37]$$

Although Eqs. [35] through [37] completely describe the response of the circuit in Fig. 8.39b to the pulse waveform of Fig. 8.39a, the current waveform itself is sensitive to both the circuit time constant τ and the voltage pulse duration t_0 . Two possible curves are shown in Fig. 8.40.

The left curve is drawn for the case where the time constant is only one-half as large as the length of the applied pulse; the rising portion of the exponential has therefore almost reached V_0/R before the decaying exponential begins. The opposite situation is shown to the right; there, the time constant is twice t_0 and the response never has a chance to reach the larger amplitudes.

The procedure we have been using to find the response of an RL circuit after dc sources have been switched on or off (or in or out of the circuit) at some instant of time is summarized in the following. We assume that the circuit is reducible to a single equivalent resistance R_{eq} in series with a single equivalent inductance L_{eq} when all independent sources are set equal to zero. The response we seek is represented by $f(t)$.

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} , L_{eq} , and the time constant $\tau = L_{eq}/R_{eq}$.
2. Viewing L_{eq} as a short circuit, use dc analysis methods to find $i_L(0^-)$, the inductor current just prior to the discontinuity.
3. Again viewing L_{eq} as a short circuit, use dc analysis methods to find the forced response. This is the value approached by $f(t)$ as $t \rightarrow \infty$; we represent it by $f(\infty)$.
4. Write the total response as the sum of the forced and natural responses: $f(t) = f(\infty) + Ae^{-t/\tau}$.
5. Find $f(0^+)$ by using the condition that $i_L(0^+) = i_L(0^-)$. If desired, L_{eq} may be replaced by a current source $i_L(0^+)$ [an open circuit if $i_L(0^+) = 0$] for this calculation. With the exception of inductor currents (and capacitor voltages), other currents and voltages in the circuit may change abruptly.
6. $f(0^+) = f(\infty) + A$ and $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-t/\tau}$, or total response = final value + (initial value – final value) $e^{-t/\tau}$.

PRACTICE

8.11 The circuit shown in Fig. 8.41 has been in the form shown for a very long time. The switch opens at $t = 0$. Find i_R at t equal to (a) 0^- ; (b) 0^+ ; (c) ∞ ; (d) 1.5 ms.

Ans: 0; 10 mA; 4 mA; 5.34 mA.

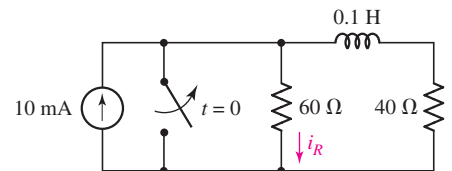


FIGURE 8.41

8.8 DRIVEN RC CIRCUITS

The complete response of any RC circuit may also be obtained as the sum of the natural and the forced response. Since the procedure is virtually identical to what we have already discussed in detail for RL circuits, the best approach at this stage is to illustrate it by working a relevant example completely, where the goal is not just a capacitor-related quantity but the current associated with a resistor as well.

EXAMPLE 8.10

Find the capacitor voltage $v_C(t)$ and the current $i(t)$ in the $200\ \Omega$ resistor of Fig. 8.42 for all time.

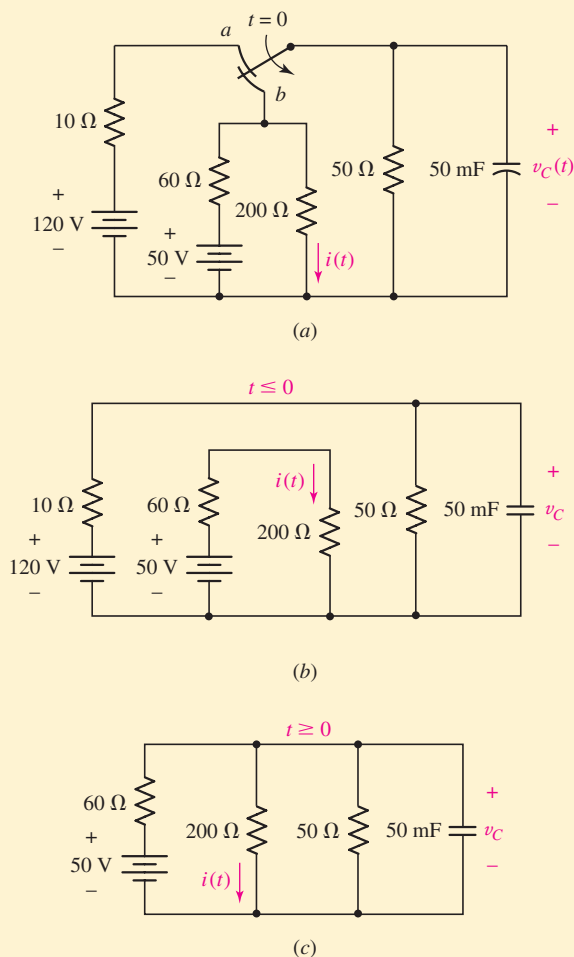


FIGURE 8.42 (a) An RC circuit in which the complete responses v_C and i are obtained by adding a forced response and a natural response. (b) Circuit for $t \leq 0$. (c) Circuit for $t \geq 0$.

We begin by considering the state of the circuit at $t < 0$, corresponding to the switch at position a as represented in Fig. 8.42b. As usual, we assume no transients are present, so that only a forced response due to the 120 V source is relevant to finding $v_C(0^-)$. Simple voltage division then gives us the initial voltage,

$$v_C(0) = \frac{50}{50 + 10}(120) = 100\text{ V}$$

Since the capacitor voltage cannot change instantaneously, this voltage is equally valid at $t = 0^-$ and $t = 0^+$.

The switch is now thrown to b , and the complete response is

$$v_C = v_{Cf} + v_{Cn}$$

The corresponding circuit has been redrawn in Fig. 8.42c for convenience. The form of the natural response is obtained by replacing the 50 V source by a short circuit and evaluating the equivalent resistance to find the time constant (in other words, we are finding the Thévenin equivalent resistance “seen” by the capacitor):

$$R_{eq} = \frac{1}{\frac{1}{50} + \frac{1}{200} + \frac{1}{60}} = 24 \, \Omega$$

Thus,

$$v_{Cn} = Ae^{-t/R_{eq}C} = Ae^{-t/1.2}$$

In order to evaluate the forced response with the switch at *b*, we wait until all the voltages and currents have stopped changing, thus treating the capacitor as an open circuit, and use voltage division once more:

$$\begin{aligned} v_{Cf} &= 50 \left(\frac{200 \parallel 50}{60 + 200 \parallel 50} \right) \\ &= 50 \left(\frac{(50)(200)/250}{60 + (50)(200)/250} \right) = 20 \, \text{V} \end{aligned}$$

Consequently,

$$v_C = 20 + Ae^{-t/1.2} \quad \text{V}$$

and from the initial condition already obtained,

$$100 = 20 + A$$

or

$$v_C = 20 + 80e^{-t/1.2} \quad \text{V} \quad t \geq 0$$

and

$$v_C = 100 \, \text{V} \quad t < 0$$

This response is sketched in Fig. 8.43a; again the natural response is seen to form a transition from the initial to the final response.

Next we attack $i(t)$. This response need not remain constant during the instant of switching. With the contact at *a*, it is evident that $i = 50/260 = 192.3$ milliamperes. When the switch moves to position *b*, the forced response for this current becomes

$$i_f = \frac{50}{60 + (50)(200)/(50 + 200)} \left(\frac{50}{50 + 200} \right) = 0.1 \, \text{ampere}$$

The form of the natural response is the same as that which we already determined for the capacitor voltage:

$$i_n = Ae^{-t/1.2}$$

Combining the forced and natural responses, we obtain

$$i = 0.1 + Ae^{-t/1.2} \quad \text{amperes}$$

(Continued on next page)

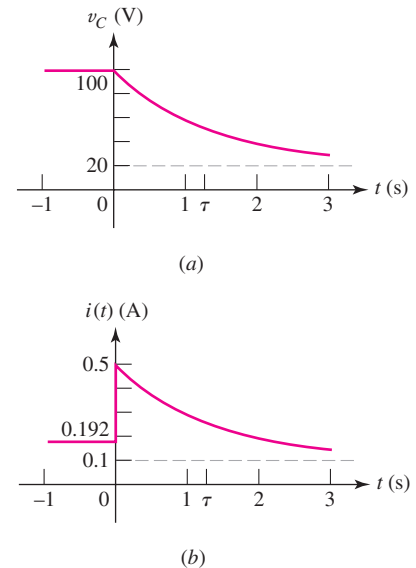


FIGURE 8.43 The responses (a) v_C and (b) i are plotted as functions of time for the circuit of Fig. 8.42.

To evaluate A , we need to know $i(0^+)$. This is found by fixing our attention on the energy-storage element (the capacitor). The fact that v_C must remain 100 V during the switching interval is the governing condition which establishes the other currents and voltages at $t = 0^+$. Since $v_C(0^+) = 100$ V, and since the capacitor is in parallel with the $200\ \Omega$ resistor, we find $i(0^+) = 0.5$ ampere, $A = 0.4$ ampere, and thus

$$i(t) = 0.1923 \text{ ampere} \quad t < 0$$

$$i(t) = 0.1 + 0.4e^{-t/1.2} \quad \text{ampere} \quad t > 0$$

or

$$i(t) = 0.1923 + (-0.0923 + 0.4e^{-t/1.2})u(t) \quad \text{amperes}$$

where the last expression is correct for all t .

The complete response for all t may also be written concisely by using $u(-t)$, which is unity for $t < 0$ and 0 for $t > 0$. Thus,

$$i(t) = 0.1923u(-t) + (0.1 + 0.4e^{-t/1.2})u(t) \quad \text{amperes}$$

This response is sketched in Fig. 8.43b. Note that only four numbers are needed to write the functional form of the response for this single-energy-storage-element circuit, or to prepare the sketch: the constant value prior to switching (0.1923 ampere), the instantaneous value just after switching (0.5 ampere), the constant forced response (0.1 ampere), and the time constant (1.2 s). The appropriate negative exponential function is then easily written or drawn.

PRACTICE

8.12 For the circuit of Fig. 8.44, find $v_C(t)$ at t equal to (a) 0^- ; (b) 0^+ ; (c) ∞ ; (d) 0.08 s.

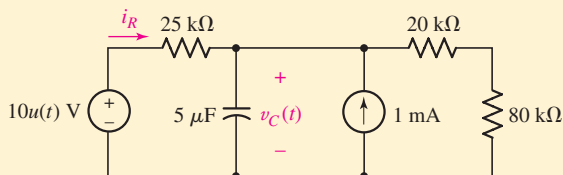


FIGURE 8.44

Ans: 20 V; 20 V; 28 V; 24.4 V.

We conclude by listing the duals of the statements given at the end of Sec. 8.7.

The procedure we have been using to find the response of an RC circuit after dc sources have been switched on or off, or in or out of the circuit, at some instant of time, say $t = 0$, is summarized in the following. We assume that the circuit is reducible to a single equivalent resistance R_{eq} in parallel with a single equivalent capacitance C_{eq} when all independent sources are set equal to zero. The response we seek is represented by $f(t)$.

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} , C_{eq} , and the time constant $\tau = R_{eq}C_{eq}$.
2. Viewing C_{eq} as an open circuit, use dc analysis methods to find $v_C(0^-)$, the capacitor voltage just prior to the discontinuity.
3. Again viewing C_{eq} as an open circuit, use dc analysis methods to find the forced response. This is the value approached by $f(t)$ as $t \rightarrow \infty$; we represent it by $f(\infty)$.
4. Write the total response as the sum of the forced and natural responses: $f(t) = f(\infty) + Ae^{-t/\tau}$.
5. Find $f(0^+)$ by using the condition that $v_C(0^+) = v_C(0^-)$. If desired, C_{eq} may be replaced by a voltage source $v_C(0^+)$ [a short circuit if $v_C(0^+) = 0$] for this calculation. With the exception of capacitor voltages (and inductor currents), other voltages and currents in the circuit may change abruptly.
6. $f(0^+) = f(\infty) + A$ and $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-t/\tau}$, or total response = final value + (initial value – final value) $e^{-t/\tau}$.

As we have just seen, the same basic steps that apply to the analysis of RL circuits can be applied to RC circuits as well. Up to now, we have confined ourselves to the analysis of circuits with dc forcing functions only, despite the fact that Eq. [30] holds for more general functions such as $Q(t) = 9 \cos(5t - 7^\circ)$ or $Q(t) = 2e^{-5t}$. Before concluding this section, we explore one such non-dc scenario.

EXAMPLE 8.11

Determine an expression for $v(t)$ in the circuit of Fig. 8.45 valid for $t > 0$.

Based on experience, we expect a complete response of the form

$$v(t) = v_f + v_n$$

where v_f will likely resemble our forcing function and v_n will have the form $Ae^{-t/\tau}$.

What is the circuit time constant τ ? We replace our source with an open circuit and find the Thévenin equivalent resistance in parallel with the capacitor:

$$R_{eq} = 4.7 + 10 = 14.7 \, \Omega$$

Thus, our time constant is $\tau = R_{eq}C = 323.4 \, \mu\text{s}$, or equivalently $1/\tau = 3.092 \times 10^3 \, \text{s}^{-1}$.

There are several ways to proceed, although perhaps the most straightforward is to perform a source transformation, resulting in a voltage source $23.5e^{-2000t}u(t)$ V in series with $14.7 \, \Omega$ and $22 \, \mu\text{F}$. (Note this does not change the time constant.)

Writing a simple KVL equation for $t > 0$, we find that

$$23.5e^{-2000t} = (14.7)(22 \times 10^{-6}) \frac{dv}{dt} + v$$

(Continued on next page)

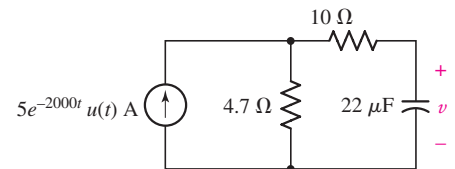
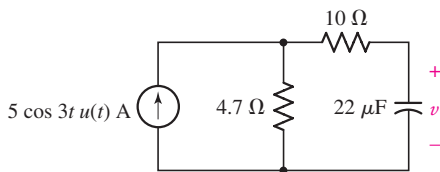


FIGURE 8.45 A simple RC circuit driven by an exponentially decaying forcing function.



■ FIGURE 8.46 A simple RC circuit driven by a sinusoidal forcing function.

A little rearranging results in

$$\frac{dv}{dt} + 3.092 \times 10^3 v = 72.67 \times 10^3 e^{-2000t}$$

which, upon comparison with Eqs. [28] and [30], allows us to write the complete response as

$$v(t) = e^{-Pt} \int Q e^{Pt} dt + A e^{-Pt}$$

where in our case $P = 1/\tau = 3.092 \times 10^3$ and $Q(t) = 72.67 \times 10^3 e^{-2000t}$. We therefore find that

$$v(t) = e^{-3092t} \int 72.67 \times 10^3 e^{-2000t} e^{3092t} dt + A e^{-3092t} \quad \text{V}$$

Performing the indicated integration,

$$v(t) = 66.55 e^{-2000t} + A e^{-3092t} \quad \text{V} \quad [38]$$

Our only source is controlled by a step function with zero value for $t < 0$, so we know that $v(0^-) = 0$. Since v is a capacitor voltage, $v(0^+) = v(0^-)$, and we therefore find our initial condition $v(0) = 0$ easily enough. Substituting this into Eq. [38], we find $A = -66.55$ V and so

$$v(t) = 66.55(e^{-2000t} - e^{-3092t}) \quad \text{V} \quad t > 0$$

PRACTICE

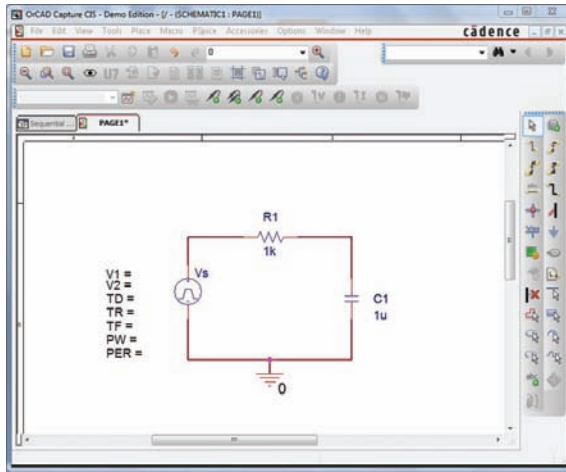
8.13 Determine the capacitor voltage v in the circuit of Fig. 8.46 for $t > 0$.

Ans: $23.5 \cos 3t + 22.8 \times 10^{-3} \sin 3t - 23.5 e^{-3092t}$ V.

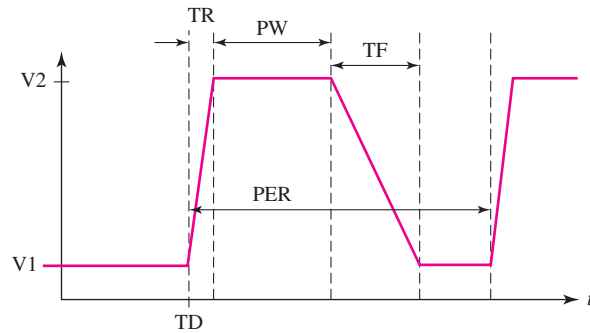
8.9 PREDICTING THE RESPONSE OF SEQUENTIALLY SWITCHED CIRCUITS

In Example 8.9 we briefly considered the response of an RL circuit to a pulse waveform, in which a source was effectively switched into and subsequently switched out of the circuit. This type of situation is common in practice, as few circuits are designed to be energized only once (passenger vehicle airbag triggering circuits, for example). In predicting the response of simple RL and RC circuits subjected to pulses and series of pulses—sometimes referred to as *sequentially switched circuits*—the key is the relative size of the circuit time constant to the various times that define the pulse sequence. The underlying principle behind the analysis will be whether the energy storage element has time to fully charge before the pulse ends, and whether it has time to fully discharge before the next pulse begins.

Consider the circuit shown in Fig. 8.47a, which is connected to a pulsed voltage source described by seven separate parameters defined in Fig. 8.47b. The waveform is bounded by two values, **V1** and **V2**. The time t_r required to change from **V1** to **V2** is called the *rise time (TR)*, and the time t_f required to change from **V2** to **V1** is called the *fall time (TF)*. The duration W_p of the pulse is referred to as the *pulse width (PW)*, and the *period T* of the waveform (**PER**) is the time it takes for the pulse to repeat. Note also that SPICE allows a time delay (**TD**) before the pulse train



(a)



(b)

FIGURE 8.47 (a) Schematic of a simple RC circuit connected to a pulsed voltage waveform. (b) Diagram of the SPICE VPULSE parameter definitions.

begins, which can be useful in allowing initial transient responses to decay for some circuit configurations.

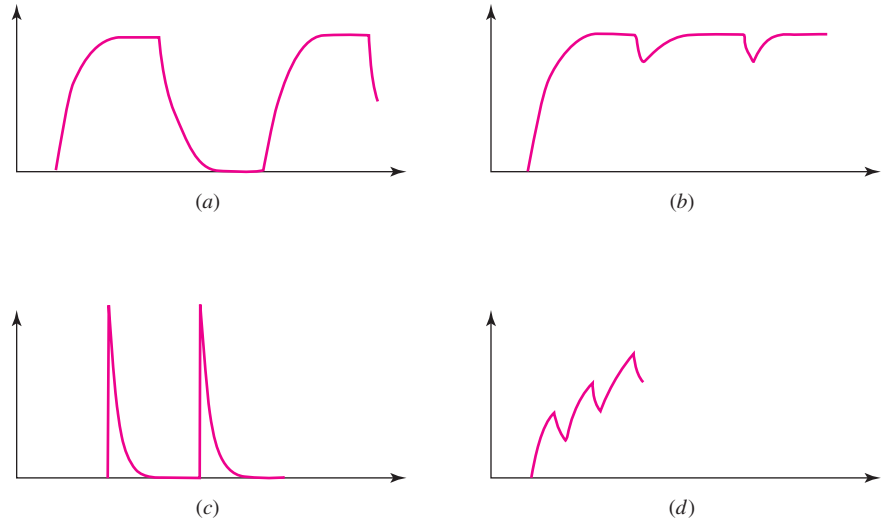
For the purposes of this discussion, we set a zero time delay, $V1 = 0$, and $V2 = 9\text{ V}$. The circuit time constant is $\tau = RC = 1\text{ ms}$, so we set the rise and fall times to be 1 ns . Although SPICE will not allow a voltage to change in zero time since it solves the differential equations using discrete time intervals, compared to our circuit time constant 1 ns is a reasonable approximation to “instantaneous.”

We will consider four basic cases, summarized in Table 8.1. In the first two cases, the pulse width W_p is much longer than the circuit time constant τ , so we expect the transients resulting from the beginning of the pulse to die out before the pulse is over. In the latter two cases, the opposite is true: the pulse width is so short that the capacitor does not have time to fully charge before the pulse ends. A similar issue arises when we consider the response of the circuit when the time between pulses ($T - W_p$) is either short (Case II) or long (Case III) compared to the circuit time constant.

TABLE 8.1 Four Separate Cases of Pulse Width and Period Relative to the Circuit Time Constant of 1 ms

Case	Pulse Width W_p	Period T
I	$10\text{ ms } (\tau \ll W_p)$	$20\text{ ms } (\tau \ll T - W_p)$
II	$10\text{ ms } (\tau \ll W_p)$	$10.1\text{ ms } (\tau \gg T - W_p)$
III	$0.1\text{ ms } (\tau \gg W_p)$	$10.1\text{ ms } (\tau \ll T - W_p)$
IV	$0.1\text{ ms } (\tau \gg W_p)$	$0.2\text{ ms } (\tau \gg T - W_p)$

We qualitatively sketch the circuit response for each of the four cases in Fig. 8.48, arbitrarily selecting the capacitor voltage as the quantity of interest as any voltage or current is expected to have the same time dependence.



■ FIGURE 8.48 Capacitor voltage for the RC circuit, with pulse width and period as in (a) Case I; (b) Case II; (c) Case III; and (d) Case IV.

In Case I, the capacitor has time to both fully charge and fully discharge (Fig. 8.48a), whereas in Case II (Fig. 8.48b), when the time between pulses is reduced, it no longer has time to fully discharge. In contrast, the capacitor does not have time to fully charge in either Case III (Fig. 8.48c) or Case IV (Fig. 8.48d).

Case I: Time Enough to Fully Charge and Fully Discharge

We can obtain exact values for the response in each case, of course, by performing a series of analyses. We consider Case I first. Since the capacitor has time to fully charge, the forced response will correspond to the 9 V dc driving voltage. The complete response to the first pulse is therefore

$$v_C(t) = 9 + Ae^{-1000t} \quad \text{V}$$

With $v_C(0) = 0$, $A = -9$ V and so

$$v_C(t) = 9(1 - e^{-1000t}) \quad \text{V} \quad [39]$$

in the interval of $0 < t < 10$ ms. At $t = 10$ ms, the source drops suddenly to 0 V, and the capacitor begins to discharge through the resistor. In this time interval we are faced with a simple “source-free” RC circuit, and we can write the response as

$$v_C(t) = Be^{-1000(t-0.01)} \quad 10 < t < 20 \text{ ms} \quad [40]$$

where $B = 8.99959$ V is found by substituting $t = 10$ ms in Eq. [39]; we will be pragmatic here and round this to 9 V, noting that the value calculated is consistent with our assumption that the initial transient dissipates before the pulse ends.

At $t = 20$ ms, the voltage source jumps immediately back to 9 V. The capacitor voltage just prior to this event is given by substituting $t = 20$ ms in Eq. [40], leading to $v_C(20 \text{ ms}) = 408.6 \mu\text{V}$, essentially zero compared to the peak value of 9 V.

If we keep to our convention of rounding to four significant digits, the capacitor voltage at the beginning of the second pulse is zero, which is the same as our starting point. Thus, Eqs. [39] and [40] form the basis of the response for all subsequent pulses, and we may write

$$v_C(t) = \begin{cases} 9(1 - e^{-1000t}) \text{ V} & 0 \leq t \leq 10 \text{ ms} \\ 9e^{-1000(t-0.01)} \text{ V} & 10 < t \leq 20 \text{ ms} \\ 9(1 - e^{-1000(t-0.02)}) \text{ V} & 20 < t \leq 30 \text{ ms} \\ 9e^{-1000(t-0.03)} \text{ V} & 30 < t \leq 40 \text{ ms} \end{cases}$$

and so on.

Case II: Time Enough to Fully Charge But Not Fully Discharge

Next we consider what happens if the capacitor is not allowed to completely discharge (Case II). Equation [39] still describes the situation in the interval of $0 < t < 10 \text{ ms}$, and Eq. [40] describes the capacitor voltage in the interval between pulses, which has been reduced to $10 < t < 10.1 \text{ ms}$.

Just prior to the onset of the second pulse at $t = 10.1 \text{ ms}$, v_C is now 8.144 V ; the capacitor has only had 0.1 ms to discharge, and therefore still retains 82 percent of its maximum energy when the next pulse begins. Thus, in the next interval,

$$v_C(t) = 9 + Ce^{-1000(t-10.1 \times 10^{-3})} \quad \text{V} \quad 10.1 < t < 20.1 \text{ ms}$$

where $v_C(10.1 \text{ ms}) = 9 + C = 8.144 \text{ V}$, so $C = -0.856 \text{ V}$ and

$$v_C(t) = 9 - 0.856e^{-1000(t-10.1 \times 10^{-3})} \quad \text{V} \quad 10.1 < t < 20.1 \text{ ms}$$

which reaches the peak value of 9 V much more quickly than for the previous pulse.

Case III: No Time to Fully Charge But Time to Fully Discharge

What if it isn't clear that the transient will dissipate before the end of the voltage pulse? In fact, this situation arises in Case III. Just as we wrote for Case I,

$$v_C(t) = 9 + Ae^{-1000t} \quad \text{V} \quad [41]$$

still applies to this situation, but now only in the interval $0 < t < 0.1 \text{ ms}$. Our initial condition has not changed, so $A = -9 \text{ V}$ as before. Now, however, just before this first pulse ends at $t = 0.1 \text{ ms}$, we find that $v_C = 0.8565 \text{ V}$. This is a far cry from the maximum of 9 V possible if we allow the capacitor time to fully charge, and is a direct result of the pulse lasting only one-tenth of the circuit time constant.

The capacitor now begins to discharge, so that

$$v_C(t) = Be^{-1000(t-1 \times 10^{-4})} \quad \text{V} \quad 0.1 < t < 10.1 \text{ ms} \quad [42]$$

We have already determined that $v_C(0.1^- \text{ ms}) = 0.8565 \text{ V}$, so $v_C(0.1^+ \text{ ms}) = 0.8565 \text{ V}$ and substitution into Eq. [42] yields $B = 0.8565 \text{ V}$. Just prior to the onset of the second pulse at $t = 10.1 \text{ ms}$, the capacitor voltage has decayed to essentially 0 V ; this is the initial condition at the start of the second pulse and so Eq. [41] can be rewritten as

$$v_C(t) = 9 - 9e^{-1000(t-10.1 \times 10^{-3})} \quad \text{V} \quad 10.1 < t < 10.2 \text{ ms} \quad [43]$$

to describe the corresponding response.

Case IV: No Time to Fully Charge or Even Fully Discharge

In the last case, we consider the situation where the pulse width and period are so short that the capacitor can neither fully charge nor fully discharge in any one period. Based on experience, we can write

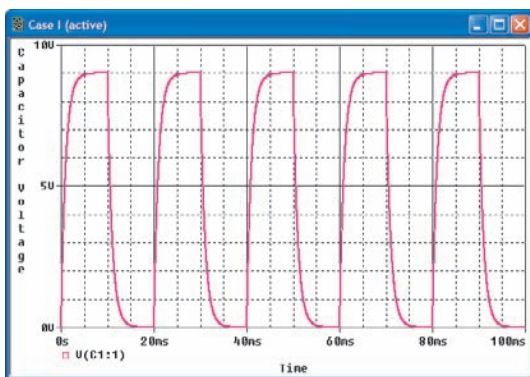
$$v_C(t) = 9 - 9e^{-1000t} \quad \text{V} \quad 0 < t < 0.1 \text{ ms} \quad [44]$$

$$v_C(t) = 0.8565e^{-1000(t-1 \times 10^{-4})} \quad \text{V} \quad 0.1 < t < 0.2 \text{ ms} \quad [45]$$

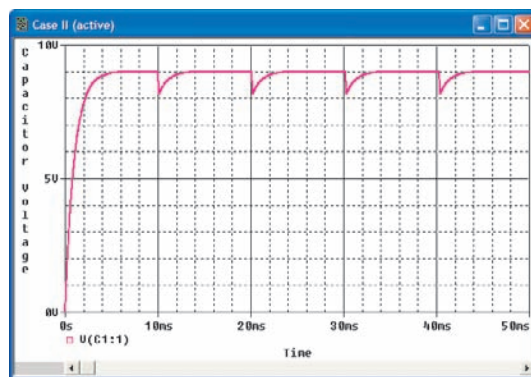
$$v_C(t) = 9 + Ce^{-1000(t-2 \times 10^{-4})} \quad \text{V} \quad 0.2 < t < 0.3 \text{ ms} \quad [46]$$

$$v_C(t) = De^{-1000(t-3 \times 10^{-4})} \quad \text{V} \quad 0.3 < t < 0.4 \text{ ms} \quad [47]$$

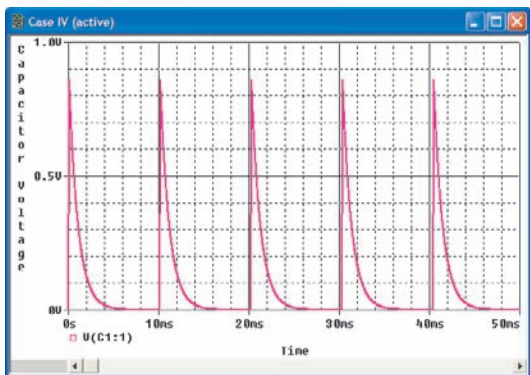
Just prior to the onset of the second pulse at $t = 0.2 \text{ ms}$, the capacitor voltage has decayed to $v_C = 0.7750 \text{ V}$; with insufficient time to fully discharge, it retains a large fraction of the little energy it had time to store initially. For the interval of $0.2 < t < 0.3 \text{ ms}$, substitution of $v_C(0.2^+) = v_C(0.2^-) = 0.7750 \text{ V}$ into Eq. [46] yields $C = -8.225 \text{ V}$. Continuing, we evaluate Eq. [46] at $t = 0.3 \text{ ms}$ and calculate $v_C = 1.558 \text{ V}$ just prior to the end of the second pulse. Thus, $D = 1.558 \text{ V}$ and our



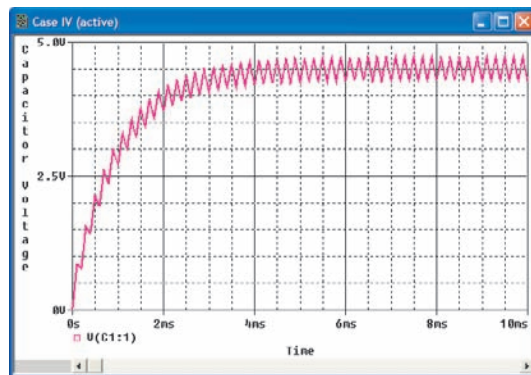
(a)



(b)



(c)



(d)

■ FIGURE 8.49 PSPice simulation results corresponding to (a) Case I; (b) Case II; (c) Case III; (d) Case IV.

capacitor is slowly charging to ever increase voltage levels over several pulses. At this stage it might be useful if we plot the detailed responses, so we show the PSpice simulation results of Cases I through IV in Fig. 8.49. Note in particular that in Fig. 8.49d, the small charge/discharge transient response similar in shape to that shown in Fig. 8.49a–c is superimposed on a charging-type response of the form $(1 - e^{-t/\tau})$. Thus, it takes about 3 to 5 circuit time constants for the capacitor to charge to its maximum value in situations where a single period does not allow it to fully charge or discharge!

What we have not yet done is predict the behavior of the response for $t \gg 5\tau$, although we would be interested in doing so, especially if it was not necessary to consider a very long sequence of pulses one at a time. We note that the response of Fig. 8.49d has an *average* value of 4.50 V from about 4 ms onward. This is exactly half the value we would expect if the voltage source pulse width allowed the capacitor to fully charge. In fact, this long-term average value can be computed by multiplying the dc capacitor voltage by the ratio of the pulse width to the period.



PRACTICE

8.14 With regard to Fig. 8.50a, sketch $i_L(t)$ in the range of $0 < t < 6$ s for (a) $v_S(t) = 3u(t) - 3u(t - 2) + 3u(t - 4) - 3u(t - 6) + \dots$; (b) $v_S(t) = 3u(t) - 3u(t - 2) + 3u(t - 2.1) - 3u(t - 4.1) + \dots$.

Ans: See Fig. 8.50b; see Fig. 8.50c.

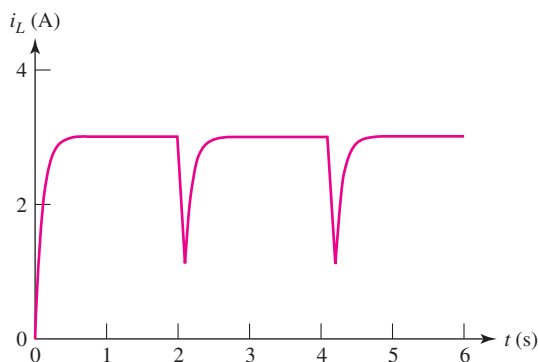
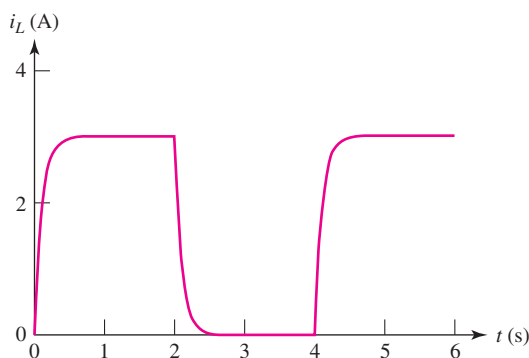
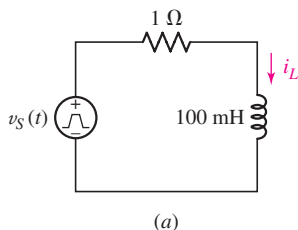
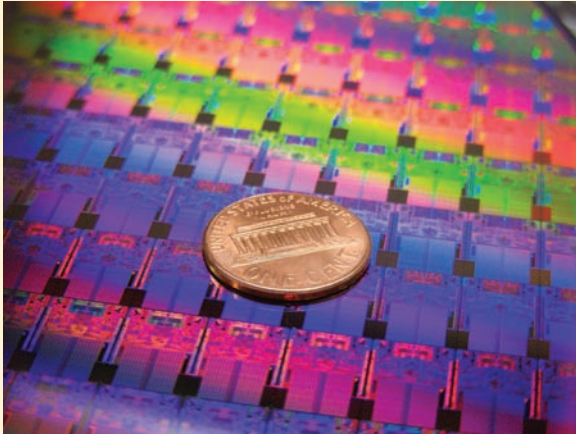


FIGURE 8.50 (a) Circuit for Practice Problem 8.14; (b) solution to part (a); (c) solution to part (b).

PRACTICAL APPLICATION

Frequency Limits in Digital Integrated Circuits

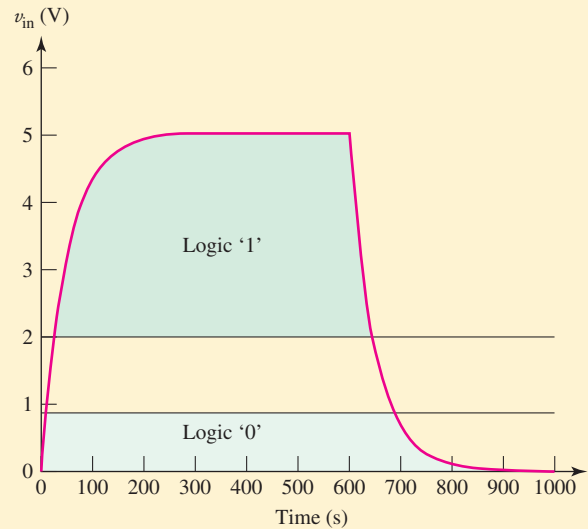
Modern digital integrated circuits such as programmable array logic (PALs) and microprocessors (Fig. 8.51) are composed of interconnected transistor circuits known as *gates*.



■ FIGURE 8.51 A silicon wafer with multiple, identical integrated circuit dies. Each die is smaller than a US 1-cent coin. Reprinted with permission of Intel Corporation.

Digital signals are represented symbolically by combinations of ones and zeros, and can be either data or instructions (such as “add” or “subtract”). Electrically, we represent a logic “1” by a “high” voltage, and a logic “0” by a “low” voltage. In practice, there is a range of voltages that correspond to each; for example, in the 7400 series of TTL logic integrated circuits, any voltage

between 2 and 5 V will be interpreted as a logic “1,” and any voltage between 0 and 0.8 V will be interpreted as a logic “0.” Voltages between 0.8 and 2 V do not correspond to either logic state, as shown in Fig. 8.52.



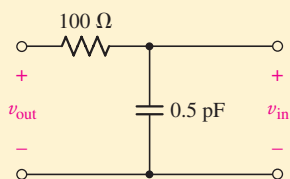
■ FIGURE 8.52 Charge/discharge characteristic of a pathway capacitance identifying the TTL voltage ranges for logic “1” and logic “0,” respectively.

A key parameter in digital circuits is the speed at which we can effectively use them. In this sense, “speed” refers to how quickly we can switch a gate from one logic state to another (either logic “0” to logic “1” or vice

SUMMARY AND REVIEW

In this chapter we learned that circuits containing a single energy storage element (either an inductor or a capacitor) can be described by a characteristic time scale, namely, the *circuit time constant* ($\tau = L/R$, or $\tau = RC$, respectively). If we attempt to change the amount of energy stored in the element (either charging or discharging), *every* voltage and current in the circuit will include an exponential term of the form $e^{-t/\tau}$. After approximately *5 time constants* from the moment we attempted to alter the amount of stored energy, the *transient* response has *essentially disappeared* and we are left simply with a *forced* response which arises from the independent sources driving the circuit at times $t > 0$. When determining the forced response in a purely dc circuit, we may treat inductors as short circuits and capacitors as open circuits.

versa), and the time delay required to convey the output of one gate to the input of the next gate. Although transistors contain “built-in” capacitances that affect their switching speed, it is the *interconnect pathways* that presently limit the speed of the fastest digital integrated circuits. We can model the interconnect pathway between two logic gates using a simple RC circuit (although as feature sizes continue to decrease in modern designs, more detailed models are required to accurately predict circuit performance). For example, consider a $2000\text{ }\mu\text{m}$ long pathway $2\text{ }\mu\text{m}$ wide. We can model this pathway in a typical silicon-based integrated circuit as having a capacitance of 0.5 pF and a resistance of $100\text{ }\Omega$, shown schematically in Fig. 8.53.



■ FIGURE 8.53 Circuit model for an integrated circuit pathway.

Let's assume the voltage v_{out} represents the output voltage of a gate that is changing from a logic “0” state to a logic “1” state. The voltage v_{in} appears across the input of a second gate, and we are interested in how long it takes v_{in} to reach the same value as v_{out} .

Assuming the 0.5 pF capacitance that characterizes the interconnect pathway is initially discharged [i.e., $v_{\text{in}}(0) = 0$], calculating the RC time constant for our

pathway as $\tau = RC = 50\text{ ps}$, and defining $t = 0$ as when v_{out} changes, we obtain the expression

$$v_{\text{in}}(t) = Ae^{-t/\tau} + v_{\text{out}}(0)$$

Setting $v_{\text{in}}(0) = 0$, we find that $A = -v_{\text{out}}(0)$ so that

$$v_{\text{in}}(t) = v_{\text{out}}(0)[1 - e^{-t/\tau}]$$

Upon examining this equation, we see that v_{in} will reach the value $v_{\text{out}}(0)$ after $\sim 5\tau$ or 250 ps . If the voltage v_{out} changes again before this transient time period is over, then the capacitance does not have sufficient time to fully charge. In such situations, v_{in} will be less than $v_{\text{out}}(0)$. Assuming that $v_{\text{out}}(0)$ equals the minimum logic “1” voltage, for example, this means that v_{in} will not correspond to a logic “1.” If v_{out} now suddenly changes to 0 V (logic “0”), the capacitance will begin to discharge so that v_{in} decreases further. Thus, by switching our logic states too quickly, we are unable to transfer the information from one gate to another.

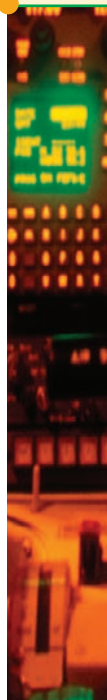
The fastest speed at which we can change logic states is therefore $(5\tau)^{-1}$. This can be expressed in terms of the maximum operating frequency:

$$f_{\text{max}} = \frac{1}{2(5\tau)} = 2\text{ GHz}$$

where the factor of 2 represents a charge/discharge period. If we desire to operate our integrated circuit at a higher frequency so that calculations can be performed faster, we need to reduce the interconnect capacitance and/or the interconnect resistance.

We started our analysis with so-called source-free circuits to introduce the idea of time constants without unnecessary distractions; such circuits have zero forced response and a transient response derived entirely from the energy stored at $t = 0$. We reasoned that a capacitor cannot change its voltage in zero time (or an infinite current results), and indicated this by introducing the notation $v_C(0^+) = v_C(0^-)$. Similarly, the current through an inductor cannot change in zero time, or $i_L(0^+) = i_L(0^-)$. The *complete* response is always the sum of the transient response and the forced response. Applying the initial condition to the complete response allows us to determine the unknown constant which multiplies the transient term.

We spent a little time discussing modeling switches, both analytically and within the context of PSpice. A common mathematical representation makes use of the unit-step function $u(t - t_0)$, which has zero value for $t < t_0$, unity value for $t > t_0$, and is indeterminate for $t = t_0$. Unit-step



functions can “activate” a circuit (connecting sources so current can flow) for values of t preceding a specific time as well as after. Combinations of step functions can be used to create pulses and more complex waveforms. In the case of sequentially switched circuits, where sources are connected and disconnected repeatedly, we found the behavior of the circuits to depend strongly on both period and pulse width as they compare to the circuit time constant.

This is a good time to highlight some key points worth reviewing, along with relevant example(s).

- ❑ The response of a circuit having sources suddenly switched in or out of a circuit containing capacitors and inductors will always be composed of two parts: a *natural* response and a *forced* response.
- ❑ The form of the natural response (also referred to as the *transient response*) depends only on the component values and the way they are wired together. (Examples 8.1, 8.2)
- ❑ A circuit reduced to a single equivalent capacitance C and a single equivalent resistance R will have a natural response given by $v(t) = V_0 e^{-t/\tau}$, where $\tau = RC$ is the circuit time constant. (Examples 8.3, 8.5)
- ❑ A circuit reduced to a single equivalent inductance L and a single equivalent resistance R will have a natural response given by $i(t) = I_0 e^{-t/\tau}$, where $\tau = L/R$ is the circuit time constant. (Example 8.4)
- ❑ Circuits with dependent sources can be represented by a resistance using Thévenin procedures.
- ❑ The unit-step function is a useful way to model the closing or opening of a switch, provided we are careful to keep an eye on the initial conditions. (Examples 8.7, 8.9)
- ❑ The form of the forced response mirrors the form of the forcing function. Therefore, a dc forcing function always leads to a constant forced response. (Examples 8.7, 8.8)
- ❑ The *complete* response of an *RL* or *RC* circuit excited by a dc source will have the form $f(0^+) = f(\infty) + A$ and $f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-t/\tau}$, or total response = final value + (initial value – final value) $e^{-t/\tau}$. (Examples 8.9, 8.10, 8.11)
- ❑ The complete response for an *RL* or *RC* circuit may also be determined by writing a single differential equation for the quantity of interest and solving. (Examples 8.2, 8.11)
- ❑ When dealing with sequentially switched circuits, or circuits connected to pulsed waveforms, the relevant issue is whether the energy storage element has sufficient time to fully charge and to fully discharge, as measured relative to the circuit time constant.

READING FURTHER

A guide to solution techniques for differential equations can be found in:

W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations and Boundary Value Problems*, 7th ed. New York: Wiley, 2002.

A detailed description of transients in electric circuits is given in:

E. Weber, *Linear Transient Analysis Volume I*. New York: Wiley, 1954.
(Out of print, but in many university libraries.)

EXERCISES

8.1 The Source-Free RL Circuit

- Setting $R = 1 \text{ k}\Omega$ and $L = 1 \text{ nH}$ for the circuit represented in Fig. 8.1, and with the knowledge that $i(0) = -3 \text{ mA}$, (a) write an expression for $i(t)$ valid for all $t \geq 0$; (b) compute $i(t)$ at $t = 0$, $t = 1 \text{ ps}$, 2 ps , and 5 ps ; and (c) calculate the energy stored in the inductor at $t = 0$, $t = 1 \text{ ps}$, and $t = 5 \text{ ps}$.
- If $i(0) = 1 \text{ A}$ and $R = 100 \Omega$ for the circuit of Fig. 8.1, (a) select L such that $i(50 \text{ ms}) = 368 \text{ mA}$; (b) compute the energy stored in the inductor at $t = 0$, 50 ms , 100 ms , and 150 ms .
- Referring to the circuit shown in Fig. 8.1, select values for both elements such that $L/R = 1$ and (a) calculate $v_R(t)$ at $t = 0$, 1 , 2 , 3 , 4 , and 5 s ; (b) compute the power dissipated in the resistor at $t = 0$, 1 s , and 5 s . (c) At $t = 5 \text{ s}$, what is the percentage of the initial energy still stored in the inductor?
- The circuit depicted in Fig. 8.1 is constructed from components whose value is unknown. If a current $i(0)$ of $6 \mu\text{A}$ initially flows through the inductor, and it is determined that $i(1 \text{ ms}) = 2.207 \mu\text{A}$, calculate the ratio of R to L .
- Determine the characteristic equation of each of the following differential equations:

$$(a) 5v + 14 \frac{dv}{dt} = 0; \quad (b) -9 \frac{di}{dt} - 18i = 0;$$

$$(c) \frac{di}{dt} + 18i + \frac{R}{B}i = 0; \quad (d) \frac{d^2f}{dt^2} + 8 \frac{df}{dt} + 2f = 0.$$

- For the following characteristic equations, write corresponding differential equations and find all roots, whether real, imaginary, or complex:
(a) $4s + 9 = 0$; (b) $2s - 4 = 0$; (c) $s^2 + 7s + 1 = 0$; (d) $5s^2 + 8s + 18 = 0$.
- With the assumption that the switch in the circuit of Fig. 8.54 has been closed a long, long, long time, calculate $i_L(t)$ at (a) the instant just before the switch opens; (b) the instant just after the switch opens; (c) $t = 15.8 \mu\text{s}$; (d) $t = 31.5 \mu\text{s}$; (e) $t = 78.8 \mu\text{s}$.

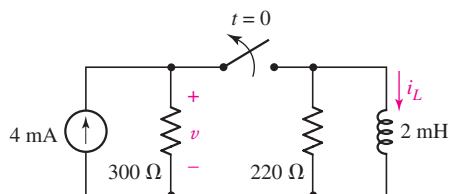


FIGURE 8.54

- The switch in Fig. 8.54 has been closed since Catfish Hunter last pitched for the New York Yankees. Calculate the voltage labeled v as well as the energy stored in the inductor at (a) the instant just prior to the switch being thrown open; (b) the instant just after the switch is opened; (c) $t = 8 \mu\text{s}$; (d) $t = 80 \mu\text{s}$.
- The switch in the circuit of Fig. 8.55 has been closed a ridiculously long time before suddenly being thrown open at $t = 0$. (a) Obtain expressions for i_L and v in the circuit of Fig. 8.55 which are valid for all $t \geq 0$. (b) Calculate $i_L(t)$ and

$v(t)$ at the instant just prior to the switch opening, at the instant just after the switch opening, and at $t = 470 \mu\text{s}$.

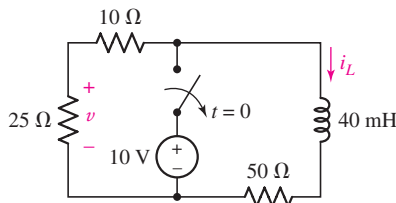


FIGURE 8.55

10. Assuming the switch initially has been open for a really, really long time, (a) obtain an expression for i_W in the circuit of Fig. 8.56 which is valid for all $t \geq 0$; (b) calculate i_W at $t = 0$ and $t = 1.3 \text{ ns}$.

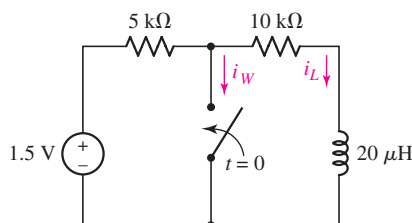


FIGURE 8.56

8.2 Properties of the Exponential Response

11. (a) Graph the function $f(t) = 10e^{-2t}$ over the range of $0 \leq t \leq 2.5 \text{ s}$ using linear scales for both y and x axes. (b) Replot with a logarithmic scale for the y axis. [Hint: the function `semilogy()` can be helpful here.] (c) What are the units of the 2 in the argument of the exponential? (d) At what time does the function reach a value of 9? 8? 1?
12. The current $i(t)$ flowing through a 1Ω resistor is given by $i(t) = 5e^{-10t} \text{ mA}$, $t \geq 0$. (a) Determine the values of t for which the resistor voltage magnitude is equal to 5 V, 2.5 V, 0.5 V, and 5 mV. (b) Graph the function over the range of $0 \leq t \leq 1 \text{ s}$ using linear scales for both axes. (c) Draw a tangent to your curve at $t = 100 \text{ ms}$, and determine where the tangent intersects the time axis.
13. The thickness of a solar cell must be chosen carefully to ensure photons are properly absorbed; even metals can be partly transparent when rolled out into very thin foils. If the incident light flux (number of photons per unit area per unit time) at the solar cell surface ($x = 0$) is given by Φ_0 , and the intensity of light a distance x inside the solar cell is given by $\Phi(x)$, the behavior of $\Phi(x)$ is described by the equation $d\Phi/dx + \alpha\Phi = 0$. Here, α , known as the absorption coefficient, is a constant specific to a given semiconductor material. (a) What is the SI unit for α ? (b) Obtain an expression for $\Phi(x)$ in terms of Φ_0 , α , and x . (c) How thick should the solar cell be made in order to absorb at least 38% of the incident light? Express your answer in terms of α . (d) What happens to the light which enters the solar cell at $x = 0$ but is not absorbed?
14. For the circuit of Fig. 8.5, compute the time constant if the 10Ω resistor is replaced with (a) a short circuit; (b) a 1Ω resistor; (c) a series connection of two 5Ω resistors; (d) a 100Ω resistor. (e) Verify your answers with a suitable parameter sweep simulation. (Hint: the cursor tool might come in handy, and the answer does not depend on the initial current you choose for the inductor.)
15. Design a circuit which will produce a voltage of 1 V at some initial time, and a voltage of 368 mV at a time 5 seconds later. You may specify an initial inductor current without showing how it arises.



8.3 The Source-Free RC Circuit

16. The resistor in the circuit of Fig. 8.57 has been included to model the dielectric layer separating the plates of the 3.1 nF capacitor, and has a value of 55 M Ω . The capacitor is storing 200 mJ of energy just prior to $t = 0$. (a) Write an expression for $v(t)$ valid for $t \geq 0$. (b) Compute the energy remaining in the capacitor at $t = 170$ ms. (c) Graph $v(t)$ over the range of $0 < t < 850$ ms, and identify the value of $v(t)$ when $t = 2\tau$.

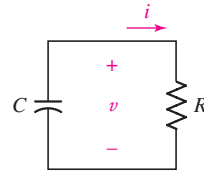


FIGURE 8.57

17. The resistor in the circuit of Fig. 8.57 has a value of 1 Ω and is connected to a 22 mF capacitor. The capacitor dielectric has infinite resistance, and the device is storing 891 mJ of energy just prior to $t = 0$. (a) Write an expression for $v(t)$ valid for $t \geq 0$. (b) Compute the energy remaining in the capacitor at $t = 11$ ms and 33 ms. (c) If it is determined that the capacitor dielectric is much leakier than expected, having a resistance as low as 100 k Ω , repeat parts (a) and (b).



18. Calculate the time constant of the circuit depicted in Fig. 8.57 if $C = 10$ mF and R is equal to (a) 1 Ω ; (b) 10 Ω ; (c) 100 Ω . (d) Verify your answers with an appropriate parameter sweep simulation. (Hint: the cursor tool might come in handy, and the time constant does not depend on the initial voltage across the capacitor.)



19. Design a capacitor-based circuit that will provide (a) a voltage of 9 V at some time $t = 0$, and a voltage of 1.2 V at a time 4 ms later; (b) a current of 1 mA at some time $t = 0$, and a reduced current of 50 μ A at a time 100 ns later. (You can choose to design two separate circuits if desired, and do not need to show how the initial capacitor voltage is set.)

20. It is safe to assume that the switch drawn in the circuit of Fig. 8.58 has been closed such a long time that any transients which might have arisen from first connecting the voltage source have disappeared. (a) Determine the circuit time constant. (b) Calculate the voltage $v(t)$ at $t = \tau$, 2τ , and 5τ .

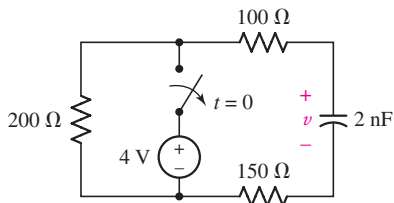


FIGURE 8.58

21. We can safely assume the switch in the circuit of Fig. 8.59 was closed a very long time prior to being thrown open at $t = 0$. (a) Determine the circuit time constant. (b) Obtain an expression for $i_1(t)$ which is valid for $t > 0$. (c) Determine the power dissipated by the 12 Ω resistor at $t = 500$ ms.

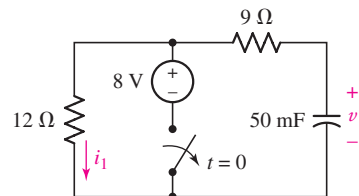


FIGURE 8.59

22. The switch above the 12 V source in the circuit of Fig. 8.60 has been closed since just after the wheel was invented. It is finally thrown open at $t = 0$. (a) Compute the circuit time constant. (b) Obtain an expression for $v(t)$ valid for $t > 0$. (c) Calculate the energy stored in the capacitor 170 ms after the switch is opened.

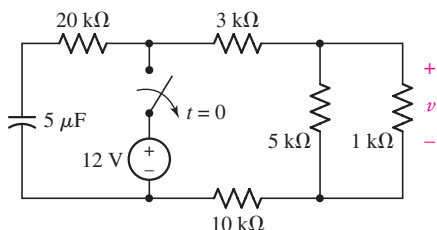


FIGURE 8.60

23. For the circuit represented schematically in Fig. 8.61, (a) calculate $v(t)$ at $t = 0$, $t = 984$ s, and $t = 1236$ s; (b) determine the energy still stored in the capacitor at $t = 100$ s.

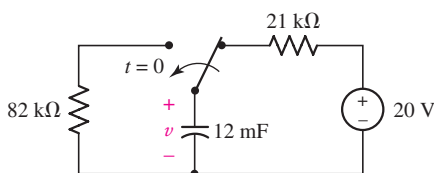


FIGURE 8.61

24. For the circuit depicted in Fig. 8.62, (a) compute the circuit time constant; (b) determine v in the instant just before the switch is closed; (c) obtain an expression for $v(t)$ valid for $t > 0$; (d) calculate $v(3$ ms).

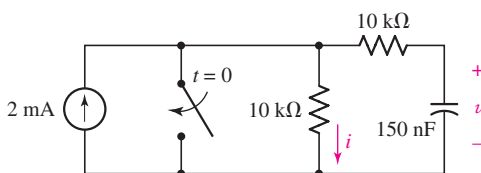


FIGURE 8.62

25. The switch drawn in Fig. 8.62 has been open a ponderously long time. (a) Determine the value of the current labeled i just prior to the switch being closed. (b) Obtain the value of i just after the switch is closed. (c) Compute the power dissipated in each resistor over the range of $0 < t < 15$ ms. (d) Graph your answer to part (c).

8.4 A More General Perspective

26. (a) Obtain an expression for $v(t)$, the voltage which appears across resistor R_3 in the circuit of Fig. 8.63, which is valid for $t > 0$. (b) If $R_1 = 2R_2 = 3R_3 = 4R_4 = 1.2$ kΩ, $L = 1$ mH, and $i_L(0^-) = 3$ mA, calculate $v(t = 500$ ns).

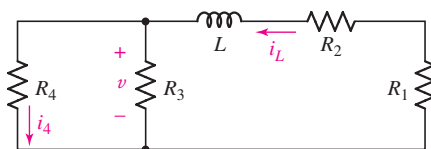


FIGURE 8.63

27. For the circuit of Fig. 8.64, determine i_x , i_L , and v_L at t equal to (a) 0^- ; (b) 0^+ .

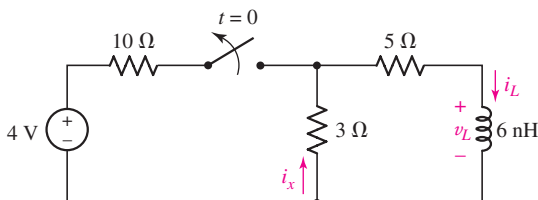


FIGURE 8.64

28. The switch shown in Fig. 8.65 has been closed for 6 years prior to being flipped open at $t = 0$. Determine i_L , v_L , and v_R at t equal to (a) 0^- ; (b) 0^+ ; (c) $1\ \mu\text{s}$; (d) $10\ \mu\text{s}$.

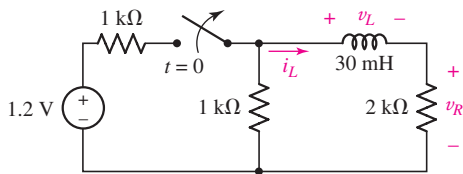


FIGURE 8.65

29. Obtain expressions for both $i_1(t)$ and $i_L(t)$ as labeled in Fig. 8.66, which are valid for $t > 0$.

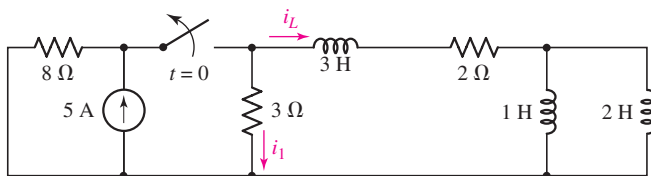


FIGURE 8.66

30. The voltage across the resistor in a simple source-free RL circuit is given by $5e^{-90t}\text{ V}$, $t > 0$. The inductor value is not known. (a) At what time will the inductor voltage be exactly one-half of its maximum value? (b) At what time will the inductor current reach 10% of its maximum value?
31. Referring to Fig. 8.67, calculate the currents i_1 and i_2 at t equal to (a) 1 ms; (b) 3 ms.

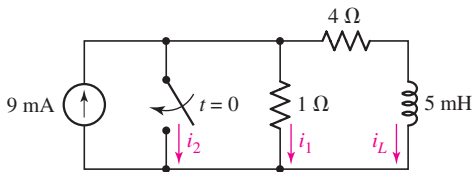


FIGURE 8.67

32. (a) Obtain an expression for v_x as labeled in the circuit of Fig. 8.68. (b) Evaluate v_x at $t = 5\text{ ms}$. (c) Verify your answer with an appropriate PSpice simulation. (Hint: employ the part named **Sw_tClose**.)

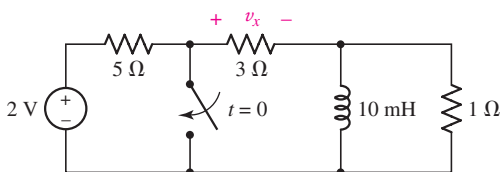


FIGURE 8.68

33. Design a complete circuit which provides a voltage v_{ab} across two terminals labeled a and b , respectively, such that $v_{ab} = 5\text{ V}$ at $t = 0^-$, 2 V at $t = 1\text{ s}$, and less than 60 mV at $t = 5$. Verify the operation of your circuit using an appropriate PSpice simulation. (Hint: employ the part named **Sw_tOpen** or **Sw_tClose** as appropriate.)



34. For the part **Sw tOpen**, PSpice actually employs a sequence of simulations where the part is first replaced with a resistor having value $1\text{ M}\Omega$, and then replaced with a resistor having value $10\text{ m}\Omega$ corresponding to when the switch opens. (a) Evaluate the reliability of these default values by simulating the circuit of Fig. 8.55, and evaluating i_L at $t = 1\text{ ns}$. (b) Repeat part (a) with **RCLOSED** changed to $1\text{ }\Omega$. Did this change your answer? (c) Repeat part (a) with **ROPEN** changed to $100\text{ k}\Omega$ and **RCLOSED** reset to its default value. Did this change your answer? (*Hint*: double-click on the part to access its attributes.)
35. Select values for the resistors R_0 and R_1 in the circuit of Fig. 8.69 such that $v_C(0.65) = 5.22\text{ V}$ and $v_C(2.21) = 1\text{ V}$.

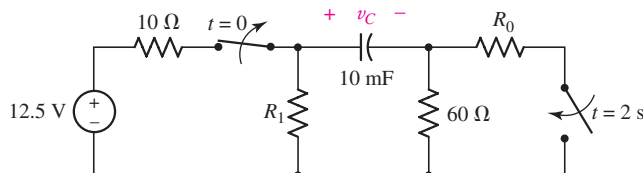


FIGURE 8.69

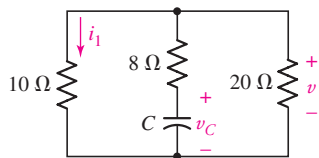


FIGURE 8.70

36. A quick measurement determines that the capacitor voltage v_C in the circuit of Fig. 8.70 is 2.5 V at $t = 0^-$. (a) Determine $v_C(0^+)$, $i_1(0^+)$, and $v(0^+)$. (b) Select a value of C so that the circuit time constant is equal to 14 s .
37. Determine $v_C(t)$ and $v_o(t)$ as labeled in the circuit represented by Fig. 8.71 for t equal to (a) 0^- ; (b) 0^+ ; (c) 10 ms ; (d) 12 ms .

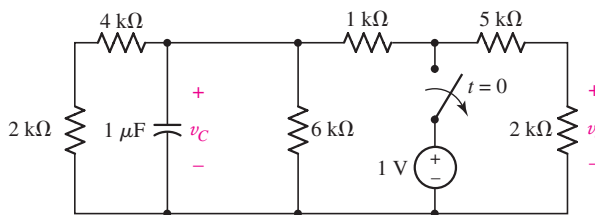


FIGURE 8.71

38. For the circuit shown in Fig. 8.72, determine (a) $v_C(0^-)$; (b) $v_C(0^+)$; (c) the circuit time constant; (d) $v_C(3\text{ ms})$.

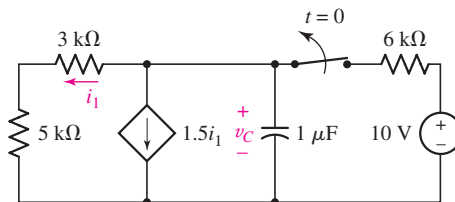


FIGURE 8.72

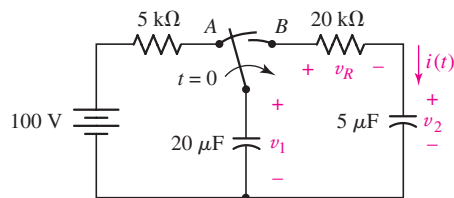


FIGURE 8.73

39. The switch in Fig. 8.73 is moved from A to B at $t = 0$ after being at A for a long time. This places the two capacitors in series, thus allowing equal and opposite dc voltages to be trapped on the capacitors. (a) Determine $v_1(0^-)$, $v_2(0^-)$, and $v_R(0^-)$. (b) Find $v_1(0^+)$, $v_2(0^+)$, and $v_R(0^+)$. (c) Determine the time constant of $v_R(t)$. (d) Find $v_R(t)$, $t > 0$. (e) Find $i(t)$. (f) Find $v_1(t)$ and $v_2(t)$ from $i(t)$ and the initial values. (g) Show that the stored energy at $t = \infty$ plus the total energy dissipated in the $20\text{ k}\Omega$ resistor is equal to the energy stored in the capacitors at $t = 0$.

40. The inductor in Fig. 8.74 is storing 54 nJ at $t = 0^-$. Compute the energy remaining at t equal to (a) 0^+ ; (b) 1 ms; (c) 5 ms.

8.5 The Unit-Step Function

41. Evaluate the following functions at $t = -2, 0$, and $+2$: (a) $f(t) = 3u(t)$; (b) $g(t) = 5u(-t) + 3$; (c) $h(t) = 5u(t - 3)$; (d) $z(t) = 7u(1 - t) + 4u(t + 3)$.
42. Evaluate the following functions at $t = -1, 0$, and $+3$: (a) $f(t) = tu(1 - t)$; (b) $g(t) = 8 + 2u(2 - t)$; (c) $h(t) = u(t + 1) - u(t - 1) + u(t + 2) - u(t - 4)$; (d) $z(t) = 1 + u(3 - t) + u(t - 2)$.
43. Sketch the following functions over the range $-3 \leq t \leq 3$: (a) $v(t) = 3 - u(2 - t) - 2u(t)$ V; (b) $i(t) = u(t) - u(t - 0.5) + u(t - 1) - u(t - 1.5) + u(t - 2) - u(t - 2.5)$ A; (c) $q(t) = 8u(-t)$ C.
44. Use step functions to construct an equation that describes the waveform sketched in Fig. 8.75.

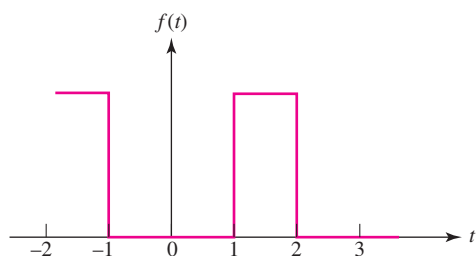


FIGURE 8.75

45. Employing step functions as appropriate, describe the voltage waveform graphed in Fig. 8.76.

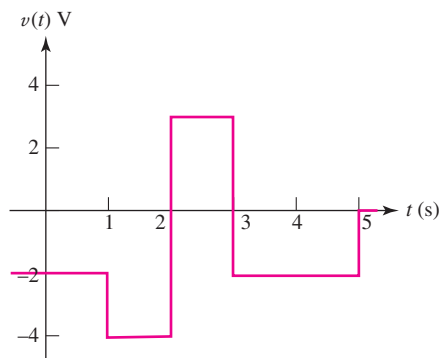


FIGURE 8.76

8.6 Driven RL Circuits

46. With reference to the simple circuit depicted in Fig. 8.77, compute $i(t)$ for (a) $t = 0^-$; (b) $t = 0^+$; (c) $t = 1^-$; (d) $t = 1^+$; (e) $t = 2$ ms.

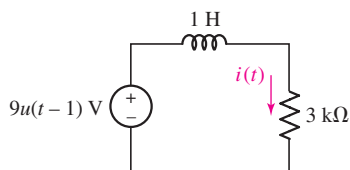


FIGURE 8.77

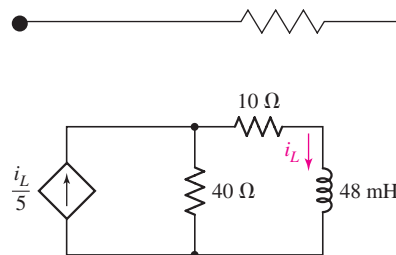


FIGURE 8.74



47. For the circuit given in Fig. 8.78, (a) determine $v_L(0^-)$, $v_L(0^+)$, $i_L(0^-)$, and $i_L(0^+)$; (b) calculate $i_L(150 \text{ ns})$. (c) Verify your answer to part (b) with an appropriate PSpice simulation.

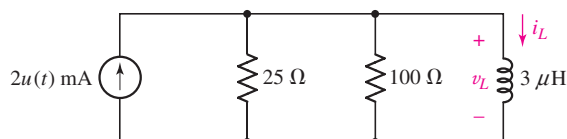


FIGURE 8.78

48. The circuit depicted in Fig. 8.79 contains two independent sources, one of which is only active for $t > 0$. (a) Obtain an expression for $i_L(t)$ valid for all t ; (b) calculate $i_L(t)$ at $t = 10 \mu\text{s}$, $20 \mu\text{s}$, and $50 \mu\text{s}$.

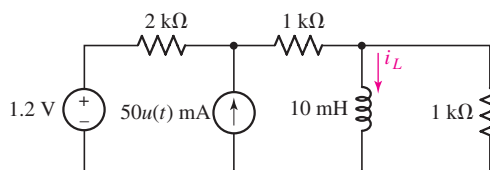


FIGURE 8.79

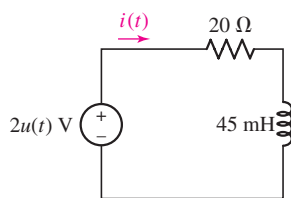


FIGURE 8.80

49. The circuit shown in Fig. 8.80 is powered by a source which is inactive for $t < 0$. (a) Obtain an expression for $i(t)$ valid for all t . (b) Graph your answer over the range of $-1 \text{ ms} \leq t \leq 10 \text{ ms}$.

50. For the circuit shown in Fig. 8.81, (a) obtain an expression for $i(t)$ valid for all time; (b) obtain an expression for $v_R(t)$ valid for all time; and (c) graph both $i(t)$ and $v_R(t)$ over the range of $-1 \text{ s} \leq t \leq 6 \text{ s}$.

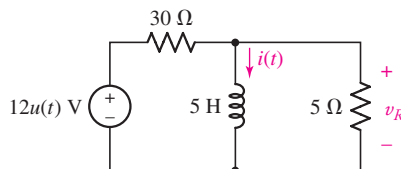


FIGURE 8.81

8.7 Natural and Forced Response

51. For the two-source circuit of Fig. 8.82, note that one source is always on. (a) Obtain an expression for $i(t)$ valid for all t ; (b) determine at what time the energy stored in the inductor reaches 99% of its maximum value.
52. (a) Obtain an expression for i_L as labeled in Fig. 8.83 which is valid for all values of t . (b) Sketch your result over the range $-1 \text{ ms} \leq t \leq 3 \text{ ms}$.

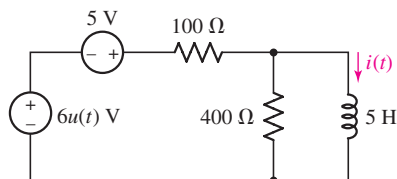


FIGURE 8.82

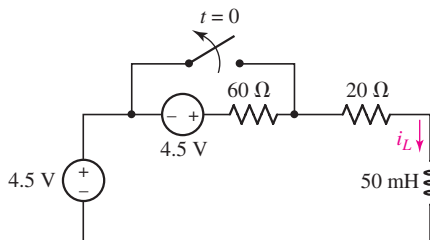


FIGURE 8.83

53. Obtain an expression for $i(t)$ as labeled in the circuit diagram of Fig. 8.84, and determine the power being dissipated in the $40\ \Omega$ resistor at $t = 2.5\text{ ms}$.

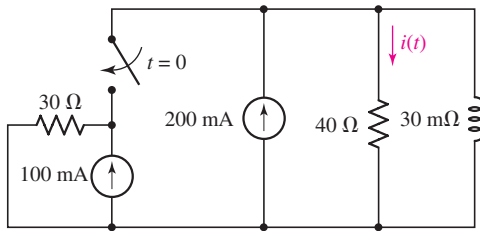


FIGURE 8.84

54. Obtain an expression for i_1 as indicated in Fig. 8.85 that is valid for all values of t .

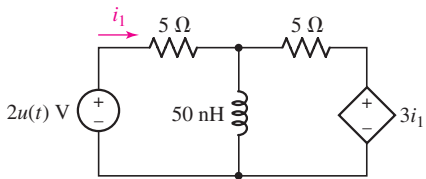


FIGURE 8.85

55. Plot the current $i(t)$ in Fig. 8.86 if (a) $R = 10\ \Omega$; (b) $R = 1\ \Omega$. In which case does the inductor (temporarily) store the most energy? *Explain.*

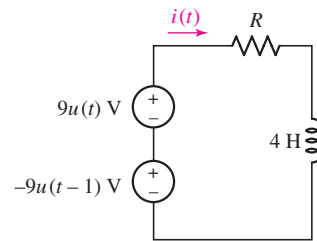


FIGURE 8.86

8.8 Driven RC Circuits

56. (a) Obtain an expression for v_C in the circuit of Fig. 8.87 valid for all values of t . (b) Sketch $v_C(t)$ over the range $0 \leq t \leq 4\ \mu\text{s}$.

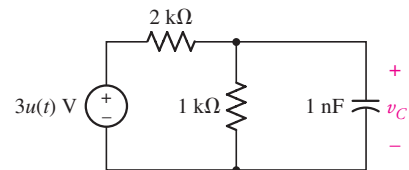


FIGURE 8.87

57. Obtain an equation which describes the behavior of i_A as labeled in Fig. 8.88 over the range of $-1\text{ ms} \leq t \leq 5\text{ ms}$.

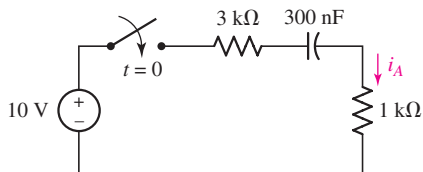


FIGURE 8.88



58. The switch in the circuit of Fig. 8.89 has been closed an incredibly long time, before being thrown open at $t = 0$. (a) Evaluate the current labeled i_x at $t = 70\text{ ms}$. (b) Verify your answer with an appropriate PSpice simulation.

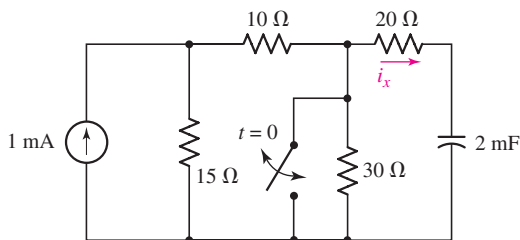


FIGURE 8.89



59. The switch in the circuit of Fig. 8.89 has been open a really, really incredibly long time, before being closed without further fanfare at $t = 0$. (a) Evaluate the current labeled i_x at $t = 70$ ms. (b) Verify your answer with an appropriate PSpice simulation.
60. The “make-before-break” switch shown in Fig. 8.90 has been in position a since the first episode of “Jonny Quest” aired on television. It is moved to position b , finally, at time $t = 0$. (a) Obtain expressions for $i(t)$ and $v_C(t)$ valid for all values of t . (b) Determine the energy remaining in the capacitor at $t = 33 \mu\text{s}$.

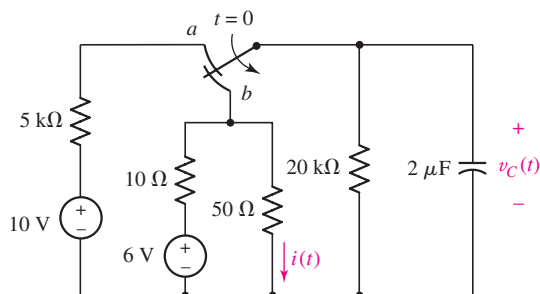


FIGURE 8.90

61. The switch in the circuit of Fig. 8.91, often called a *make-before-break* switch (since during switching it briefly makes contact to both parts of the circuit to ensure a smooth electrical transition), moves to position b at $t = 0$ only after being in position a long enough to ensure all initial transients arising from turning on the sources have long since decayed. (a) Determine the power dissipated by the 5Ω resistor at $t = 0^-$. (b) Determine the power dissipated in the 3Ω resistor at $t = 2$ ms.

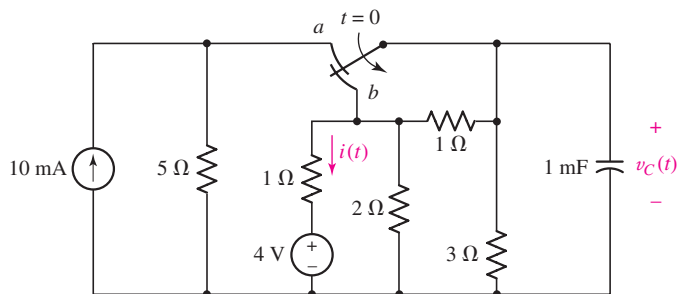


FIGURE 8.91

62. Referring to the circuit represented in Fig. 8.92, (a) obtain an equation which describes v_C valid for all values of t ; (b) determine the energy remaining in the capacitor at $t = 0^+$, $t = 25 \mu\text{s}$, and $t = 150 \mu\text{s}$.

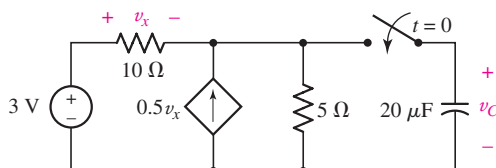


FIGURE 8.92

63. The dependent source shown in Fig. 8.92 is unfortunately installed upside down during manufacturing, so that the terminal corresponding to the arrowhead is actually wired to the negative reference terminal of the voltage source. This is not detected by the quality assurance team so the unit ships out wired improperly. The capacitor is initially discharged. If the $5\ \Omega$ resistor is only rated to $2\ \text{W}$, after what time t is the circuit likely to fail?
64. For the circuit represented in Fig. 8.93, (a) obtain an expression for v which is valid for all values of t ; (b) sketch your result for $0 \leq t \leq 3\ \text{s}$.

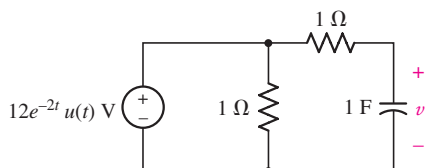


FIGURE 8.93

65. Obtain an expression for the voltage v_x as labeled in the op amp circuit of Fig. 8.94.

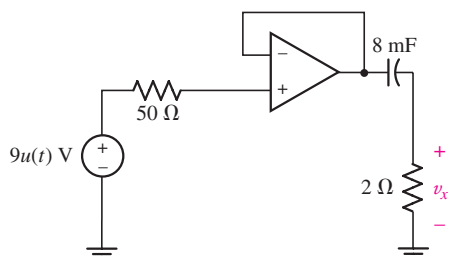


FIGURE 8.94

8.9 Predicting the Response of Sequentially Switched Circuits

66. Sketch the current i_L of the circuit in Fig. 8.50a if the $100\ \text{mH}$ inductor is replaced by a $1\ \text{nH}$ inductor, and is subjected to the waveform $v_s(t)$ equal to
 (a) $5u(t) - 5u(t - 10^{-9}) + 5u(t - 2 \times 10^{-9})\ \text{V}$, $0 \leq t \leq 4\ \text{ns}$;
 (b) $9u(t) - 5u(t - 10^{-8}) + 5u(t - 2 \times 10^{-8})\ \text{V}$, $0 \leq t \leq 40\ \text{ns}$.
67. The $100\ \text{mH}$ inductor in the circuit of Fig. 8.50a is replaced with a $1\ \text{H}$ inductor. Sketch the inductor current i_L if the source $v_s(t)$ is equal to
 (a) $5u(t) - 5u(t - 0.01) + 5u(t - 0.02)\ \text{V}$, $0 \leq t \leq 40\ \text{ms}$;
 (b) $5u(t) - 5u(t - 10) + 5u(t - 10.1)\ \text{V}$, $0 \leq t \leq 11\ \text{s}$.



68. Sketch the voltage v_C across the capacitor of Fig. 8.95 for at least 3 periods if $R = 1\ \Omega$, $C = 1\ \text{F}$, and $v_s(t)$ is a pulsed waveform having (a) minimum of $0\ \text{V}$, maximum of $2\ \text{V}$, rise and fall times of $1\ \text{ms}$, pulse width of $10\ \text{s}$, and period of $10\ \text{s}$; (b) minimum of $0\ \text{V}$, maximum of $2\ \text{V}$, rise and fall times of $1\ \text{ms}$, pulse width of $10\ \text{ms}$, and period of $10\ \text{ms}$. (c) Verify your answers with appropriate PSpice simulations.



69. Sketch the voltage v_C across the capacitor of Fig. 8.95 for at least 3 periods if $R = 1\ \Omega$, $C = 1\ \text{F}$, and $v_s(t)$ is a pulsed waveform having (a) minimum of $0\ \text{V}$, maximum of $2\ \text{V}$, rise and fall times of $1\ \text{ms}$, pulse width of $10\ \text{s}$, and period of $10\ \text{ms}$; (b) minimum of $0\ \text{V}$, maximum of $2\ \text{V}$, rise and fall times of $1\ \text{ms}$, pulse width of $10\ \text{ms}$, and period of $10\ \text{s}$. (c) Verify your answers with appropriate PSpice simulations.

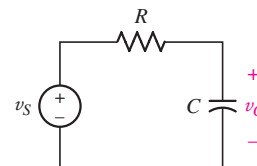


FIGURE 8.95

Chapter-Integrating Exercises

70. The circuit in Fig. 8.96 contains two switches that always move in perfect synchronization. However, when switch *A* opens, switch *B* closes, and vice versa. Switch *A* is initially open, while switch *B* is initially closed; they change positions every 40 ms. Using the bottom node as the reference node, determine the voltage across the capacitor at *t* equal to (a) 0^- ; (b) 0^+ ; (c) 40^- ms; (d) 40^+ ms; (e) 50 ms.

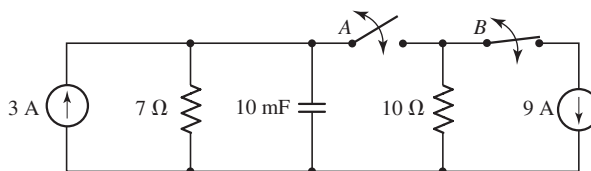


FIGURE 8.96

71. In the circuit of Fig. 8.96, when switch *A* opens, switch *B* closes, and vice versa. Switch *A* is initially open, while switch *B* is initially closed; they change positions every 400 ms. Determine the energy in the capacitor at *t* equal to (a) 0^- ; (b) 0^+ ; (c) 200 ms; (d) 400^- ms; (e) 400^+ ms; (f) 700 ms.
72. Refer to the circuit of Fig. 8.97, which contains a voltage-controlled dependent voltage source in addition to two resistors. (a) Compute the circuit time constant. (b) Obtain an expression for v_x valid for all *t*. (c) Plot the power dissipated in the resistor over the range of 6 time constants. (d) Repeat parts (a) to (c) if the dependent source is installed in the circuit upside down. (e) Are both circuit configurations “stable”? Explain.
73. In the circuit of Fig. 8.97, a 3 mF capacitor is accidentally installed instead of the inductor. Unfortunately, that’s not the end of the problems, as it’s later determined that the real capacitor is not really well modeled by an ideal capacitor, and the dielectric has a resistance of 10 kΩ (which should be viewed as connected in parallel to the ideal capacitor). (a) Compute the circuit time constant with and without taking the dielectric resistance into account. By how much does the dielectric change your answer? (b) Calculate v_x at $t = 200$ ms. Does the dielectric resistance affect your answer significantly? Explain.

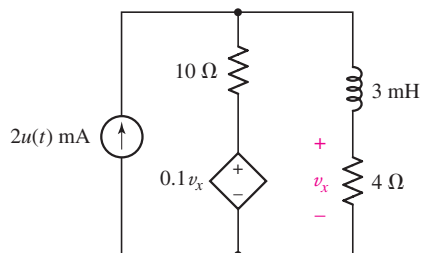


FIGURE 8.97

74. For the circuit of Fig. 8.98, assuming an ideal op amp, derive an expression for $v_o(t)$ if v_s is equal to (a) $4u(t)$ V; (b) $4e^{-130,000t}u(t)$ V.

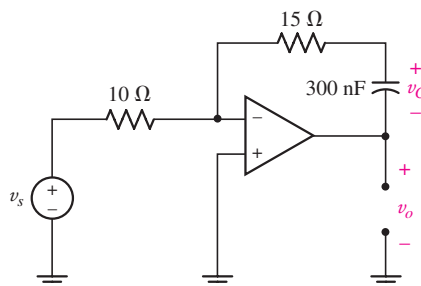


FIGURE 8.98